**A Dissertation on**

# Predicting next day movement of Indian stock market sectors based on geopolitical risks

*submitted in partial fulfilment of the requirement for the award of the degree of*

**MASTER OF SCIENCE**

in

**Computer Science (BIG DATA ANALYTICS)**

submitted by

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under the guidance of

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Department of Data Science and Analytics
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**CENTRAL UNIVERSITY OF RAJASTHAN**

**September 2026**

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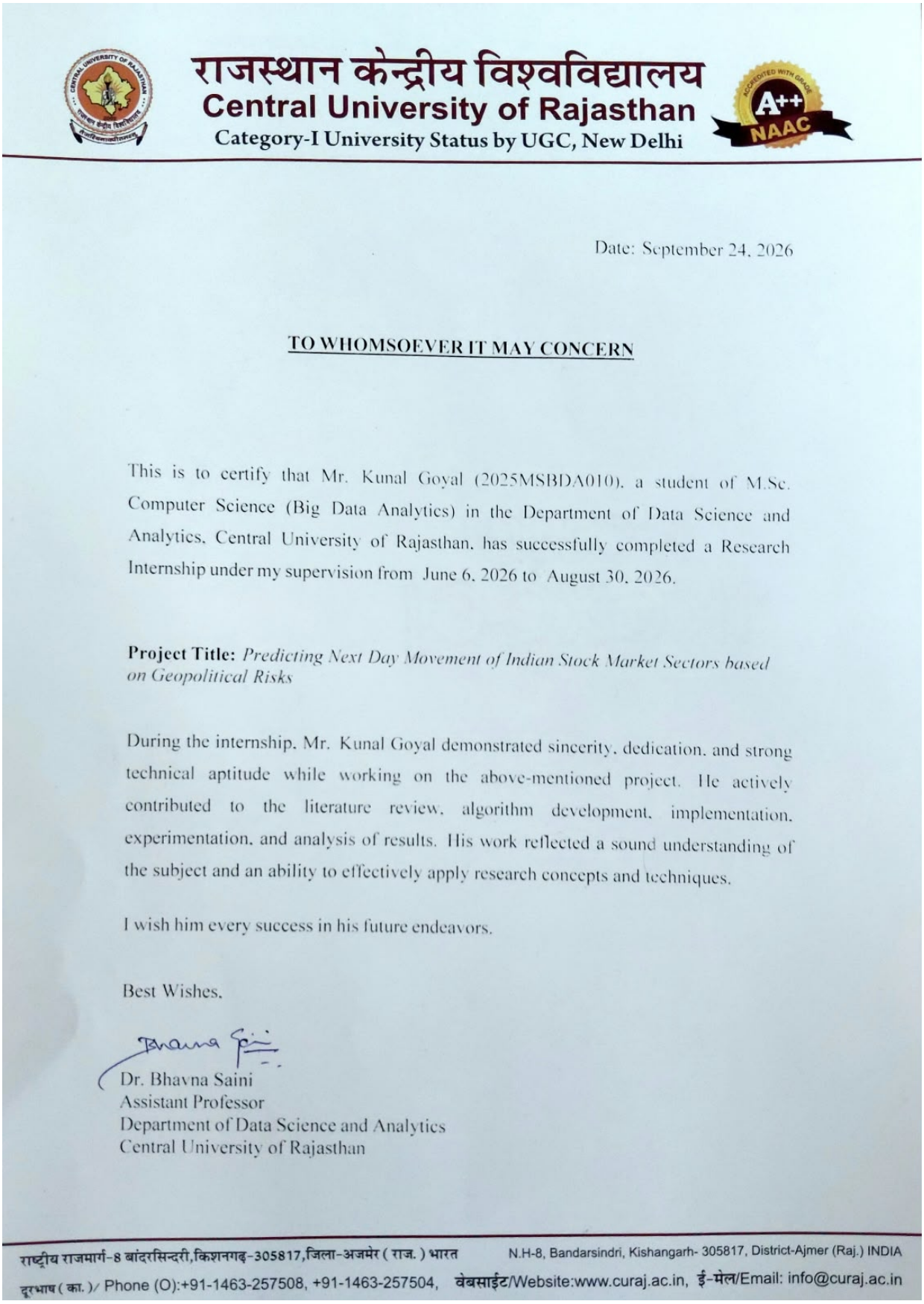
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# INTERNSHIP CERTIFICATE



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**TO WHOMSOEVER IT MAY CONCERN**

This is to certify that Mr. Kunal Goyal (2025MSBDA010), a student of M.Sc. Computer Science (Big Data Analytics) in the Department of Data Science and Analytics, Central University of Rajasthan, has successfully completed a Research Internship under my supervision from June 6, 2026 to August 30, 2026.

**Project Title:** *Predicting Next Day Movement of Indian Stock Market Sectors based on Geopolitical Risks*

During the internship, Mr. Kunal Goyal demonstrated sincerity, dedication, and strong technical aptitude while working on the above-mentioned project. He actively contributed to the literature review, algorithm development, implementation, experimentation, and analysis of results. His work reflected a sound understanding of the subject and an ability to effectively apply research concepts and techniques.

I wish him every success in his future endeavors.

Best Wishes,

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Kunal Goyal
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# ABSTRACT

Predicting the direction in which a stock-market sector will move on the next trading day is a problem with clear practical appeal and a long record of mixed empirical success. Indian NIFTY sectoral indices are shaped partly by their own price history, partly by macroeconomic conditions such as commodity prices and exchange rates, and partly by external events whose influence is difficult to quantify directly. Most machine-learning treatments of this problem represent each trading day as an independent row of features, which leaves the relationships between sectors, macroeconomic variables and geopolitical actors implicit. This dissertation examines whether making those relationships explicit, through a graph representation processed by Graph Neural Networks, offers a measurable advantage over classical machine learning for next-day directional prediction of eight NIFTY sectoral indices, and whether geopolitical information derived from the GDELT event database contributes measurable predictive value.

A dataset covering 2,086 trading days between 1 January 2018 and 19 June 2026 was reconstructed from sector price files, macroeconomic series for gold, oil, USD/INR and India VIX, and GDELT event data for seven countries together with a global aggregate. Two classical models, Random Forest and XGBoost, were trained under two feature configurations, one including GDELT-derived features (E1) and one excluding them (E2), across all eight sectors, giving 32 fitted models. In parallel, a 21-node domain-informed graph containing sector, macroeconomic, country and global GDELT nodes, connected by 56 unique undirected relationships, was constructed, and four graph architectures (GCN, GraphSAGE, GAT and an HGT-style heterogeneous attention model) were trained on it. Both pipelines used a strictly chronological train, validation and test split with all preprocessing fitted on training data only.

The strongest classical configuration by mean accuracy was XGBoost without GDELT features, reaching 51.67% accuracy, 44.23% macro F1 and 0.5134 ROC-AUC on the held-out test period. Ranked by ROC-AUC instead, Random Forest with GDELT features was marginally ahead at 0.5315, so the ablation does not support a consistent conclusion in either direction; what can be stated is that the GDELT features produced no reliable improvement under the tested configurations. Among the graph models, the HGT-style model obtained the highest ROC-AUC at 0.5018, followed by GAT (0.4936), GCN (0.4858) and GraphSAGE (0.4855), all close to or below the 0.50 random-classification level. Inspection of the confusion matrices showed that all four graph models assigned the positive (UP) class to every test observation in every sector, so their apparently high F1 of 0.6814 is an artefact of constant-UP prediction evaluated under a positive-class F1 definition rather than evidence of discrimination. These results are reported exactly as obtained; they characterise an abbreviated training configuration and should not be read as an upper bound on what these architectures can achieve.

The classical pipeline was implemented in Python 3 using scikit-learn for the Random Forest, imputation and metric computation, and the XGBoost library for the gradient-boosted models. The graph pipeline was implemented directly in PyTorch 2.10.0 on CPU, without PyTorch Geometric, with NetworkX used for graph visualisation and Matplotlib for all figures. Interpretability was examined using gradient-based feature and edge saliency rather than the GNNExplainer algorithm.

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# CHAPTER 1
# INTRODUCTION

## *1.1 Introduction*

The National Stock Exchange of India publishes a family of sectoral indices alongside its broad NIFTY 50 benchmark. Each sectoral index aggregates the price behaviour of the largest listed companies operating within a defined part of the economy, and each responds to a somewhat different mixture of domestic demand conditions, global commodity cycles, currency movements and policy decisions. Forecasting whether such an index will close higher or lower on the following trading day is a binary classification problem that is easy to state and notoriously difficult to solve. Daily equity returns carry a low signal-to-noise ratio, the efficient-market argument set out by Fama [1] implies that readily available information should already be reflected in current prices, and reported improvements over a naive baseline in the wider financial-forecasting literature are often small and can fail to survive a change of market, sample period or evaluation protocol, a pattern this dissertation returns to when the results in Chapter 5 are discussed.

This dissertation approaches the problem from a relational angle. The question is not only whether past prices and external indicators carry predictive information, but whether representing the connections between sectors, macroeconomic variables and geopolitical entities as an explicit graph allows a model to use that information more effectively than a conventional tabular representation would. The work is deliberately framed as a comparison rather than a demonstration: the objective is to establish what the experiments actually show, including the possibility that the additional structure provides no benefit.

## *1.2 Background and Motivation*

Sectors do not move independently of one another. Banking and real estate share exposure to interest-rate conditions; energy and metals respond to overlapping commodity cycles; information technology and pharmaceuticals both carry substantial foreign-currency revenue exposure. A rise in crude oil prices tends to raise input costs for automobile manufacturers while improving realisations for domestic energy producers, so the same external shock can push two sectors in opposite directions. Any model that treats the eight sectoral series as eight unrelated forecasting problems discards this structure entirely.

Standard machine-learning pipelines handle interdependence implicitly. Every variable is placed side by side in a single feature vector for each trading day, and the model is free to discover whatever statistical associations exist. It receives no prior indication of which variables are related to which, and it treats each trading day as an independent observation. A graph representation encodes that prior structure directly: sectors, macroeconomic variables and countries become nodes, plausible relationships between them become edges, and message passing allows information to flow along those edges. This kind of prior structure is often described as a relational inductive bias: the architecture is told in advance which entities are considered related, which reduces the amount of structure that must be inferred from data alone. Whether such a bias improves prediction in a specific application is an empirical question, and it is the question this work sets out to answer.

The geopolitical component is motivated by a similar line of reasoning. Events involving major oil producers or large trading partners plausibly coincide with movements in commodity prices, currency values and investor sentiment, which in turn are associated with sector performance. Caldara and Iacoviello [2] show that a newspaper-based measure of geopolitical risk is associated with lower investment and higher market volatility, which establishes that such information is at least economically meaningful

at aggregate horizons. The GDELT project [3] provides machine-coded daily counts of events, conflict activity and average media tone drawn from global news coverage, offering one quantifiable proxy for this activity at daily frequency. It is an indirect proxy: GDELT measures reported news activity rather than underlying geopolitical reality, and its usefulness for a specific short-horizon prediction task cannot be assumed in advance. That is precisely why the present study tests its contribution by ablation rather than including it on the strength of plausibility alone.

### *1.3 Problem Statement*

Given price-derived features, macroeconomic indicators and GDELT-derived geopolitical measures observable up to and including trading day t, the task is to predict whether the closing value of a given NIFTY sectoral index will be higher on day t+1 than on day t. The wider problem addressed by this dissertation is whether a graph-based relational representation of the market provides a measurable advantage over classical tabular machine learning on this task, and whether geopolitical features contribute measurably to either approach.

### *1.4 Objectives*

1. Assemble a dataset combining NIFTY sectoral price data, macroeconomic series and GDELT-derived geopolitical indicators over a common daily time base.
2. Formulate next-day sector movement as a binary classification problem evaluated under a leakage-conscious chronological protocol.
3. Establish classical machine-learning baselines using Random Forest and XGBoost across all eight sectors.
4. Evaluate the contribution of GDELT-derived features through a direct ablation between a full feature set and a no-GDELT feature set.
5. Construct a domain-informed graph linking sector, macroeconomic, country and global GDELT nodes.
6. Train and evaluate four graph neural network architectures on that graph under the same protocol.
7. Compare the classical and graph-based approaches using metrics that are computed identically in both pipelines.
8. Examine model behaviour beyond aggregate metrics, including prediction distributions, training curves and gradient-based saliency.
9. Identify the limitations of the experiments performed and specify the improvements that follow directly from them.

### *1.5 Research Questions*

The study is organised around three questions, each of which is answered from the experimental evidence in Section 5.13.

- RQ1: Does the inclusion of GDELT-derived geopolitical features improve next-day sector-direction prediction relative to an otherwise identical configuration using sector-technical features alone?
- RQ2: Does any evaluated GNN architecture outperform the strongest classical machine-learning configuration under the tested experimental protocol?

- RQ3: Do the evaluated GNN models learn a sufficiently discriminative decision boundary on the held-out test period?

### *1.6 Scope of the Study*

The study covers eight NIFTY sectoral indices: Auto, Bank, Energy, FMCG, IT, Metal, Pharma and Realty. NIFTY 50 is included as a benchmark node in the graph and as a reference series for relative-strength features, but it is not itself a prediction target. Macroeconomic coverage is limited to gold, crude oil, the USD/INR exchange rate and the India VIX. Geopolitical coverage is limited to GDELT-derived measures for the United States, India, China, Russia, Pakistan, Israel and Iran, together with a global aggregate node.

The prediction target is next-day direction only. No attempt is made to forecast return magnitude, and no trading simulation or transaction-cost analysis is performed, so the results carry no implication about whether any configuration would be profitable in practice. All conclusions are restricted to the dataset, graph topology, feature set and training configuration actually evaluated.

### *1.7 Contributions*

1. A domain-informed heterogeneous market graph with 21 nodes and 56 unique undirected relationships spanning sectoral, macroeconomic, country and global GDELT entities, with every edge group justified on economic or geopolitical grounds and explicitly presented as a modelling assumption rather than a causal claim.

2. A direct ablation of GDELT-derived features across two algorithms and eight sectors, which tests rather than assumes their contribution to a short-horizon directional task.

3. A comparison of four graph architectures against classical baselines under a single chronological protocol with training-only preprocessing, reported on metrics that are computed identically in both pipelines.

4. A transparent discussion of the near-random results, showing that the graph models did not provide a strong improvement over the baseline models for short-term sector prediction.

### *1.8 Organization of the Report*

Chapter 2 reviews the financial and technical background and the related literature, closing with the research gap that motivates the study. Chapter 3 sets out the data, feature construction, graph design and experimental protocol. Chapter 4 describes the implementation, including the software and hardware environment and the steps required for reproduction. Chapter 5 presents and discusses the results, including a diagnostic analysis of the graph models. Chapter 6 states the conclusions, limitations and future scope.

# CHAPTER 2
# BACKGROUND MATERIAL AND RELATED WORK

## *2.1 The Indian Stock Market and NIFTY Sectoral Indices*

The NIFTY sectoral indices are capitalisation-weighted aggregates of the largest listed companies within defined segments of the Indian economy. Each index therefore summarises the collective price behaviour of a group of firms exposed to broadly similar business conditions, while averaging away much of the idiosyncratic movement that affects any individual stock.

Sector-level modelling occupies a useful middle ground for a study of this kind. Individual stocks carry substantial firm-specific noise arising from earnings announcements, management changes and order-book dynamics, most of which has no relationship to macroeconomic or geopolitical conditions. A single broad index, at the other extreme, averages away precisely the differences that make sector behaviour interesting. Sectoral indices retain the distinctions that matter here: the differing exposure of each sector to commodity prices, exchange rates, interest rates and external events. The eight indices used in this work span domestic-demand sectors such as FMCG and Auto, globally exposed sectors such as IT and Pharma, commodity-linked sectors such as Energy and Metal, and interest-rate-sensitive sectors such as Bank and Realty.

## *2.2 Financial Returns and Directional Prediction*

For a sectoral index with closing price P(s,t) on trading day t, the simple return is defined in Equation 2.1 and the log return, which is additive across time and better behaved statistically, in Equation 2.2.

$$r_{s,t} = \frac{P_{s,t} - P_{s,t-1}}{P_{s,t-1}} \tag{2.1}$$

$$\ell_{s,t} = \ln\left(\frac{P_{s,t}}{P_{s,t-1}}\right) \tag{2.2}$$

Forecasting the magnitude of r(s,t+1) is a regression problem with a very low signal-to-noise ratio at daily frequency. Reducing the task to the sign of the next return converts it into binary classification, which is more tractable and is the formulation adopted throughout this dissertation. The reduction is not free of cost. A model may predict direction correctly on many small moves while being wrong on the few large ones that dominate economic outcomes, so classification accuracy alone is not a measure of economic value. This is one reason why the efficient-market position described by Fama [1] is not refuted by a modest accuracy advantage: statistical predictability and exploitable predictability are different claims, and only the first is examined here.

A further consequence of the binary formulation is that the class distribution becomes relevant. Equity indices drift upward over long samples, so the UP class is usually slightly more frequent than the DOWN class. A model that always predicts UP will therefore achieve an accuracy equal to the prevalence of the positive class without learning anything at all, which makes a constant-prediction baseline an essential reference point. Section 5.2 reports the observed prevalence for each sector, and Section 5.8 shows why this matters for interpreting the graph results.

### 2.3 Classical Machine Learning for Financial Prediction

Tree-ensemble methods are a natural starting point for tabular financial data. They handle mixed feature scales without normalisation, capture non-linear interactions between predictors, and are comparatively resistant to overfitting when regularised appropriately.

Random Forest, introduced by Breiman [4], builds a large number of decision trees on bootstrap samples of the training data, restricting each split to a random subset of the available features, and averages their predictions. The randomisation decorrelates the individual trees, which reduces the variance of the ensemble relative to a single deep tree. XGBoost, described by Chen and Guestrin [5], takes the opposite approach: trees are built sequentially, each fitted to the residual errors of its predecessors, with explicit regularisation terms penalising model complexity and with shrinkage applied to each new tree. The two methods therefore represent the two dominant ensemble strategies, bagging and boosting, which is why both are used as baselines in this study rather than a single representative.

No single tree-ensemble method is guaranteed to dominate across sample sizes and market regimes, which is one reason Random Forest and XGBoost are both retained as baselines here rather than a single method being selected in advance on the strength of prior literature; their relative performance on the present task is left to the results in Chapter 5 rather than assumed. Fischer and Krauss [6] report that long short-term memory networks outperform simpler classifiers on directional prediction for S&P 500 constituents, but also document that the advantage decays substantially in later, more efficient sample periods, which is a useful reminder that results obtained in one regime need not transfer to another.

### 2.4 Geopolitical Risk and Financial Markets

A substantial body of empirical work links geopolitical risk measures to financial market behaviour. Caldara and Iacoviello [2] construct a geopolitical risk index from the frequency of newspaper articles covering adverse geopolitical events and show that increases in the index are associated with lower investment and employment and with higher stock-market volatility. Their measure operates at monthly frequency, which is typical of the literature.

More recent work applies machine learning directly. Chowdhury et al. [7] forecast energy-market tail risk using several machine-learning models with explainable-AI attribution and report that geopolitical indicators contribute approximately 19.15% of the predictive power of their best model, alongside foreign-exchange and monetary-policy variables.

The available evidence supports the treatment of geopolitical information as structured, quantifiable model input. However, it does not establish that such features will help at a daily horizon for equity sector direction, which is a narrower, noisier and considerably harder task than tail-risk forecasting. The distinction is important: evidence that geopolitical risk matters for markets in general is not evidence that a particular geopolitical proxy will improve a particular short-horizon classifier.

### 2.5 The GDELT Event Database

GDELT, introduced by Leetaru and Schrodt [3], is an open database of machine-coded events extracted from global news media in near real time. Each event record carries attributes including the actors involved, an event classification based on the CAMEO coding scheme, geographic location, a Goldstein conflict-cooperation score and an average tone score derived from the surrounding coverage. Aggregating these records by day and by country yields time series of event counts, conflict-related activity and average tone that can be aligned with daily market data.

Two characteristics of the database matter for the present study. First, GDELT measures news coverage rather than events themselves, so shifts in media attention can register as changes in the data without any corresponding change in underlying conditions; a widely reported event and an equally significant

but under-reported one are not treated equally. Second, the coding is fully automated, which introduces classification error, duplicate reporting and occasional spurious actor attribution. GDELT is therefore best understood as an indirect and noisy proxy for geopolitical activity. This is a further reason why its contribution is tested by ablation in this work rather than assumed, and why a negative ablation result should be read as a statement about this representation of geopolitical information rather than about geopolitical information in general.

### *2.6 Graph Theory Fundamentals*

A graph G = (V, E) consists of a set of nodes V and a set of edges E $\subseteq$ V $\times$ V. Node attributes are collected in a feature matrix X of shape |V| $\times$ d, and connectivity is represented by an adjacency matrix A of shape |V| $\times$ |V| whose entry A(i,j) is one when an edge joins node i and node j and zero otherwise. For an undirected graph the adjacency matrix is symmetric. The degree matrix D is the diagonal matrix whose entries are the row sums of A.

A graph is described as heterogeneous when its nodes or edges carry distinct types. The graph constructed in this dissertation is heterogeneous in both senses: it contains four node types (sector, macroeconomic, country and global) connected by six qualitatively different relation types. Heterogeneity is not merely descriptive. It affects which architectures are appropriate, because a model that treats all neighbours identically cannot distinguish a sector-to-sector relationship from a country-to-sector relationship even though the two encode very different economic assumptions.

### *2.7 Graph Neural Networks*

Graph Neural Networks extend the convolution operation to graph-structured data through message passing. At each layer, a node aggregates representations from its neighbours, combines the aggregate with its own current representation, and produces an updated representation. After L layers, a node representation reflects information drawn from its L-hop neighbourhood. The resulting family of architectures differs mainly in how neighbour messages are weighted and combined, as the specific architectures reviewed below illustrate.

The appeal for financial modelling follows from the relational inductive bias discussed in Section 1.2: the graph supplies prior information about which entities are considered related, so the model does not have to infer that structure from data alone. Whether the bias helps depends on two conditions that are not guaranteed. The assumed relationships must be informative for the specific target, and enough training data must exist for the model to exploit them. If either condition fails, the additional structure simply adds parameters and constraints without adding usable information, and the graph model may perform no better than, or worse than, a tabular baseline.

### *2.8 Graph Convolutional Networks*

The Graph Convolutional Network of Kipf and Welling [8] propagates features through a symmetric normalisation of the adjacency matrix with added self-loops, as given in Equation 2.3.

$$H^{(l+1)} = \sigma\Big(\tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}H^{(l)}W^{(l)}\Big) \tag{2.3}$$

Here $\tilde{D}$ is the degree matrix of A + I, H(l) denotes the layer-l node representations, W(l) is a learnable weight matrix and σ is a non-linearity. Neighbour contributions are weighted purely by degree, so a GCN cannot learn that some neighbours matter more than others; the normalisation is fixed by the topology. In this study the GCN serves as the simplest graph baseline against which the more expressive architectures are compared.

### 2.9 GraphSAGE

GraphSAGE [9] aggregates neighbour representations and concatenates the result with the node's own previous representation before applying a learnable transformation. Compared with the blended averaging performed by a GCN, this preserves node identity more explicitly, since the node's own features enter the update through a separate set of weights. In the context of this work, the architecture was included to test whether maintaining a clearer separation between a sector's own technical features and its neighbourhood context improves prediction.

### 2.10 Graph Attention Networks

The Graph Attention Network [10] replaces fixed degree-based weighting with attention coefficients that are learned for each edge, as shown in Equation 2.4, where a is a learnable attention vector, W a shared weight matrix, h(i) the representation of node i, and N(i) the neighbourhood of node i.

$$\alpha_{ij} = \frac{\exp\big(\text{LeakyReLU}\big(a^\top [Wh_i \,\|\, Wh_j]\big)\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\text{LeakyReLU}\big(a^\top [Wh_i \,\|\, Wh_k]\big)\big)} \tag{2.4}$$

This allows the network to weight neighbours by learned relevance rather than by degree alone, and the mechanism is closely related to the scaled dot-product attention used in Transformer architectures. In a financial application the property is attractive, because the importance of a relationship such as the one assumed between Iran and the Energy sector plausibly varies with circumstances rather than being constant. GAT was therefore included to test whether attention-based aggregation offers an advantage over the fixed aggregation used by GCN and GraphSAGE.

### 2.11 Heterogeneous Graph Transformers

The Heterogeneous Graph Transformer of Hu et al. [11] conditions attention on node and edge type, so that different relation types are processed with different learned parameters rather than being collapsed into a single homogeneous neighbourhood. Because the graph constructed in this work is explicitly heterogeneous, containing four node types and six relation types, an HGT-style model was included to test whether type-aware attention captures structure that type-agnostic architectures cannot. The implementation used in this project follows the type-conditioned attention principle, using node-type embeddings and per-relation attention biases, rather than reproducing the full architecture of the original paper; it is therefore described throughout as HGT-style, and its results should not be read as a benchmark of the published model.

### 2.12 Explainability in Graph Neural Networks

Interpreting a trained graph model generally means identifying which input features and which edges most influenced a prediction. Two broad families of method exist. Perturbation-based approaches such as GNNExplainer [12] learn soft masks over the node features and the edges of a computation subgraph, optimising the masks to preserve the model output while remaining sparse; the resulting mask is a learned, model-agnostic explanation. Gradient-based approaches, following the saliency-map idea introduced by Simonyan et al. [**?** ] and refined by more sophisticated gradient-based attribution methods developed since, instead use partial derivatives of the output with respect to the inputs as a measure of local influence.

This project uses gradient-based saliency. The distinction is maintained consistently throughout the dissertation, because the two families answer subtly different questions. Gradient saliency measures the local sensitivity of the output to infinitesimal changes in the input around a specific evaluation point; it does not identify a minimal sufficient subgraph, it is not optimised for faithfulness, and it is not equivalent

to the GNNExplainer algorithm. Describing it as GNNExplainer would misstate what was computed.

### *2.13 Related Work*

Graph-based financial prediction rests on the premise that assets sit within a relational structure that a model can exploit. The main methodological differences between published studies lie in how that structure is obtained.

These studies operate at the level of individual companies, where relationships such as shareholding, supply contracts or common ownership are documented and verifiable. The present work operates at sector level, where the corresponding relationships cannot be looked up and must instead be asserted on economic reasoning, which is a weaker foundation and is treated as such throughout.

A second line combines multiple graph views or additional modalities. Review articles by Wang et al. [13] and Patel et al. [14] survey the field as a whole. Patel et al. examine graph-based stock forecasting systematically and observe that the majority of published graphs are constructed either from fixed correlation thresholds or from sector classifications, and that GCN, GAT and GraphSAGE remain the most frequently benchmarked backbones. That observation is the direct reason those three architectures, together with a heterogeneous attention model, form the comparison set in this dissertation.

Two further observations follow from the literature reviewed above and shape the present study. Within these studies, graphs are generally built from price correlations or documented corporate relationships rather than from geopolitical entities, so this review found little direct guidance on whether a sector-to-country topology is useful. In addition, among the studies examined here, reported outcomes are predominantly positive for the architecture being proposed, and none of the reviewed studies reports a graph-based model failing to beat a simpler baseline. This is stated as a property of the literature surveyed for this dissertation rather than as a claim about the field as a whole, since a review of this scope cannot establish that no such study exists; it does, however, make it harder to judge from the reviewed literature alone how reliably published gains transfer to new datasets, new markets and stricter evaluation protocols.

### *2.14 Research Gap*

Three gaps motivate this dissertation. Each is stated as a limitation of the literature reviewed above rather than as a claim about the field as a whole, since a targeted review of this size cannot establish that no such study exists.

- G1. Country-level geopolitical event data is rarely represented as graph structure connected to the sectors it is assumed to affect. Where geopolitical information appears in the reviewed studies, it is generally an additional tabular feature rather than a node in a relational structure.
- G2. The contribution of geopolitical features is frequently assumed rather than tested against an otherwise identical configuration without them, so it is difficult to separate the value of the geopolitical signal from the value of the modelling changes introduced alongside it.
- G3. Among the studies examined in this review, negative and near-random outcomes are seldom reported in detail. Understanding why a carefully constructed pipeline fails to beat a simple baseline is informative for subsequent work, and its comparative absence from the reviewed literature is treated here as a gap in the sample reviewed rather than as a settled property of the field.

This work addresses all three. It builds a sector, macroeconomic and country graph organised around GDELT signals; it tests those signals by direct ablation; and it reports the resulting near-random outcome together with a diagnostic analysis of the model behaviour that produced it.

# CHAPTER 3
# METHODOLOGY

## *3.1 Overall Research Framework*

The study runs two parallel modelling pipelines over a shared dataset and a shared chronological evaluation protocol. The classical pipeline represents each trading day as a flat feature vector and trains tree-ensemble models separately for each sector. The graph pipeline represents each trading day as a 21-node graph with node-level features and trains graph neural networks that predict all eight sectors jointly from a single forward pass. Holding the underlying data, the splitting scheme and the evaluation period constant across both pipelines is what makes the comparison meaningful; the differences that remain between them are documented explicitly in Section 3.3 and Section 3.16. Figure 3.1 shows the overall framework.

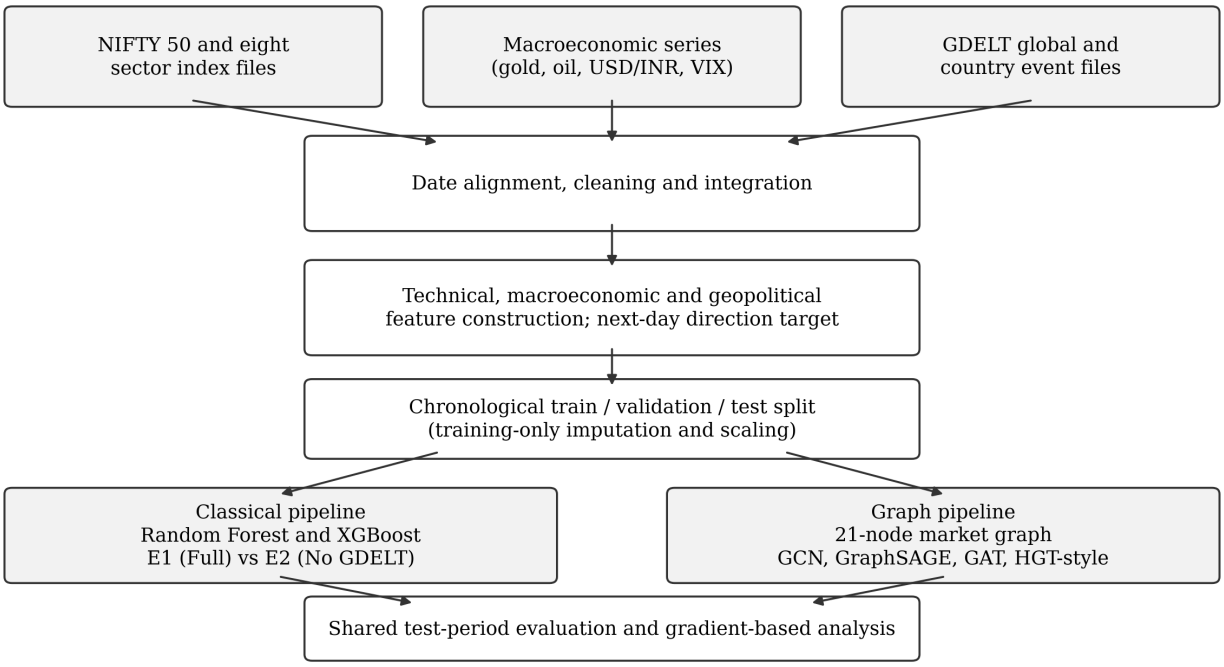



**Figure 3.1:** Overall research framework, from raw component files through shared preprocessing to the two parallel modelling pipelines and a common evaluation stage.

## *3.2 Dataset and Data Sources*

A prebuilt merged modelling table was not present in the execution environment for either notebook, so both pipelines reconstruct the modelling table from component files at run time. This is stated explicitly because it is the data path actually exercised in the executed experiments, and because it has consequences for feature coverage that are discussed in Section 3.5 and recorded as a limitation in Section 6.4.

The sources are per-sector market files for NIFTY 50 and the eight sectoral indices; macroeconomic series for gold, crude oil, the USD/INR exchange rate and the India VIX; and GDELT files providing global event aggregates together with per-country conflict, event and tone measures for the seven countries listed in Section 1.6. Table 3.1 summarises the reconstructed dataset as built by the classical pipeline.

**Table 3.1:** Summary of the reconstructed dataset (classical ML pipeline).

| Property | Value |
|---|---|
| Date range | 2018-01-01 to 2026-06-19 |
| Rows after reconstruction | 2,086 |
| Raw columns | 114 |
| Prediction targets | 8 (one per sector) |
| Sectors | Auto, Bank, Energy, FMCG, IT, Metal, Pharma, Realty |
| Benchmark series | NIFTY 50 (context and relative strength only) |
| Macroeconomic series | Gold, Oil, USD/INR, India VIX |
| GDELT countries | USA, India, China, Russia, Pakistan, Israel, Iran |

Not every collected variable is used identically in both pipelines, and this must be stated before the individual stages are described, since the distinction between a variable being collected, engineered and actually used in a given model is easy to blur in a project of this size. Table 3.2 records that distinction explicitly for every category of raw data assembled for this study.

**Table 3.2:** Collected variables and where each is actually used. A tick indicates the variable enters that model's input; a dash indicates it does not.

| Data source | Collected | Engineered | Classical ML (E1) | Classical ML (E2) | Graph / GNN |
|---|---|---|---|---|---|
| Sector prices (8 sectors) | Yes | 63 technical features | Yes | Yes | Yes (9 index nodes with NIFTY 50) |
| NIFTY 50 (benchmark) | Yes | Relative-strength input only | Indirectly (via relative strength) | Indirectly (via relative strength) | Yes (benchmark node) |
| Gold, Oil, USD/INR, India VIX | Yes | Level, return, MA30 per series | No (no dedicated macro columns in the reconstruction; see Section 3.5) | No | Yes (4 macro nodes) |
| GDELT country data (7 countries) | Yes | 21 features (conflict, event, tone) | Yes | No | Yes (7 country nodes) |
| GDELT global aggregate | Yes | 12 features | Yes | No | Yes (1 global node) |

The single most consequential entry in this table is the macroeconomic row. Gold, oil, USD/INR and India VIX were collected and are used directly by the graph pipeline to populate the four macroeconomic nodes, but no dedicated macroeconomic feature columns survive in the classical reconstruction, for the reasons given in Section 3.5. Every classical-ML claim in Chapter 5, and in particular the GDELT ablation of Section 5.4, should be read against this table rather than against the wider list of variables collected for the project as a whole.

### *3.3 Data Integration*

The market and macroeconomic files are supplied in a format carrying a two-row header, so the date column is located by matching a four-digit year, month and day pattern rather than by assuming a fixed

column position. Rows that do not match the pattern are discarded, numeric columns are coerced, duplicate dates are removed and each series is sorted before being joined on date.

The two pipelines join the component files differently, which is the reason their row counts differ, and the difference is deliberate rather than accidental. The classical pipeline performs a left join of the market table against the GDELT tables, retaining market rows even where GDELT coverage is absent, because a tree ensemble can accept an imputed value in a subset of columns. The graph pipeline performs an inner join across the sector and macroeconomic series and against the global GDELT file, so that every retained trading day has a complete set of market observations for all 21 nodes; a graph forward pass requires a full node-feature tensor and cannot be evaluated on a partially populated graph. The resulting partition sizes are given in Table 3.5 and Table 3.6.

### *3.4 Data Preprocessing*

Infinite values arising from ratio-based indicators are converted to missing values before imputation. In the classical pipeline, missing values are filled by median imputation with the median computed on the training partition only and applied unchanged to validation and test. Genuinely absent GDELT values, as distinct from technical-indicator warm-up gaps, are filled with zero on the assumption that the absence of a coded record corresponds to an absence of reported activity rather than to an unknown quantity. The graph pipeline standardises all node features using a scaler whose mean and variance are computed on the training partition only, and drops warm-up rows for which any engineered indicator is undefined.

Fitting every preprocessing step on training data alone is deliberate. Computing an imputation median or a standardisation statistic on the full series before splitting would allow distributional information from the test period to influence the training representation. Bergmeir and Benítez [15] discuss this class of problem for time-series evaluation more generally; it is easily overlooked because the resulting leakage is subtle and inflates performance without producing any obvious anomaly in the metrics.

### *3.5 Feature Engineering*

Sector-technical features are computed per series from the closing price alone, so that every predictor used for day t+1 is observable at the close of day t. Table 3.3 lists the indicators and their definitions. All rolling statistics use a minimum-periods requirement equal to the window length, so no indicator is computed from a partial window.

**Table 3.3:** Sector-technical indicators computed for each index series.

| **Indicator** | **Definition** |
|---|---|
| Level | Closing price of the index on day t |
| Return | One-day simple percentage change |
| Log return | Natural logarithm of the one-day price ratio |
| MA30 | 30-day rolling mean of the closing price |
| Volatility | 20-day rolling standard deviation of daily returns |
| Momentum | 10-day percentage change in closing price |
| Relative strength | 20-day sector return minus 20-day NIFTY 50 return |
| RSI | 14-day relative strength index |
| Drawdown | Closing price relative to its 30-day rolling maximum, minus one |
| Price/MA30 | Closing price divided by its 30-day moving average, minus one |

In the classical reconstruction this yields 63 sector-technical feature columns across the nine index series. Two feature configurations are then compared, and this comparison is the mechanism by which RQ1 is tested:

- E1 (Technical + GDELT): 63 sector-technical features, 21 country-level GDELT features and 12 global GDELT features, giving 96 features in total.
- E2 (Technical Only): the 63 sector-technical features only.

Table 3.4 sets out the feature groups. One point requires explicit statement. The classical reconstruction identified no dedicated macroeconomic feature columns distinct from the sector-technical and GDELT groups, because the macroeconomic series are not present under the expected column names in the reconstructed table. This is a direct consequence of building the modelling table from component files, and it is recorded as a limitation in Section 6.4, since it means the E1 versus E2 ablation isolates the contribution of GDELT on top of sector-technical features alone rather than on top of a fuller feature set including macroeconomic variables. The graph pipeline is not affected in the same way, because it reads the four macroeconomic series directly in order to populate its macroeconomic nodes.

**Table 3.4:** Feature groups in the classical ML pipeline.

| **Feature group** | **Count** | **Present in** |
|---|---|---|
| Sector-technical | 63 | E1 and E2 |
| GDELT country-level | 21 | E1 only |
| GDELT global | 12 | E1 only |
| Dedicated macroeconomic | 0 | — |
| E1 (Technical + GDELT) total | 96 | |
| E2 (Technical Only) total | 63 | |

### *3.6 Geopolitical Feature Construction*

GDELT-derived features enter the study at two levels. Country-level features provide a conflict count, an event count and an average tone value for each of the seven countries, giving 21 features in total. Global features provide twelve aggregate measures covering average tone, average Goldstein score, event count, total mentions, and counts of protest, threat, military, assault, fight, mass-violence, diplomacy and cooperation events.

These features describe reported news activity aggregated to a daily frequency. They are used as associative indicators of external conditions, and no claim is made anywhere in this dissertation that a change in a GDELT series causes a change in sector prices. The direction of any observed association cannot be established from the design used here, and the ablation in Section 5.4 tests only whether the features carry usable predictive information for the specific target.

### *3.7 Target Construction*

For sector s on trading day t, the binary target is defined on the next day's closing price, as given in Equation 3.1.

$$Y_{s,t+1} = \begin{cases} 1 & \text{if } P_{s,t+1} > P_{s,t} \text{ (UP)} \\ 0 & \text{otherwise (DOWN)} \end{cases} \tag{3.1}$$

A label of 1 denotes an UP day and a label of 0 denotes a DOWN day, where a flat close is treated as DOWN because the strict inequality is not satisfied. All features used to predict Y(s,t+1) are computed from information dated day t or earlier. The target is constructed before the final row of the table is dropped, since that row has no observable next-day outcome, and technical-indicator warm-up rows are removed only after all features have been computed, so that no indicator is silently shortened by an earlier deletion.

### *3.8 Data Splitting*

The data is split chronologically and never shuffled, since random splitting of a time series allows a model to be trained on observations that follow the ones it is evaluated on. Both pipelines use the same boundary dates: training ends on 31 December 2022 and validation ends on 30 June 2024, with the remainder of the sample forming the test period. Row counts differ between the pipelines because of the join and completeness differences described in Section 3.3, and because the graph pipeline additionally requires every engineered column to be defined. The two test periods therefore share the same starting date, 1 July 2024, but differ slightly in their ending dates, 19 June 2026 for the classical pipeline against 17 June 2026 for the graph pipeline, because of these data-completeness requirements; they are consequently described in this dissertation as overlapping chronological evaluation periods rather than as an identical test set. Table 3.5 and Table 3.6 give the resulting partitions, and Figure 3.2 shows them on a common time axis.

**Table 3.5:** Chronological split used by the classical ML pipeline.

| **Partition** | **Rows** | **Start** | **End** |
|---|---|---|---|
| Train | 1,234 | 2018-01-01 | 2022-12-30 |
| Validation | 365 | 2023-01-02 | 2024-06-28 |
| Test | 487 | 2024-07-01 | 2026-06-19 |

**Table 3.6:** Chronological split used by the graph (GNN) pipeline.

| **Partition** | **Rows** | **Start** | **End** |
|---|---|---|---|
| Train | 1,161 | 2018-02-16 | 2022-12-30 |
| Validation | 352 | 2023-01-03 | 2024-06-28 |
| Test | 470 | 2024-07-01 | 2026-06-17 |

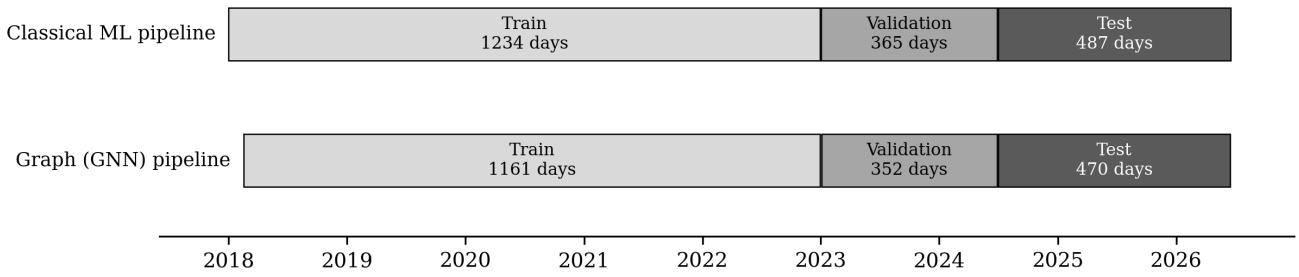



**Figure 3.2:** Chronological train, validation and test partitions for both pipelines. The two test periods share the same starting date but differ slightly in their ending dates and row counts because of the join and completeness requirements described in Section 3.3.

### *3.9 Classical Machine Learning Experimental Design*

Random Forest and XGBoost are each trained for every sector under both feature configurations, giving 2 algorithms $\times$ 2 configurations $\times$ 8 sectors = 32 fitted models. Each model is fitted on the training partition, and each is evaluated exactly once on the test partition. The hyperparameters, which are fixed in advance rather than tuned, are listed in Table 3.7. XGBoost uses the validation partition for early stopping; Random Forest has no equivalent mechanism and is fitted directly on the training partition.

**Table 3.7:** Classical model hyperparameters.

| Model | Settings |
|---|---|
| Random Forest | 500 trees; minimum 20 samples per leaf; square-root feature sampling at each split; random seed 42 |
| XGBoost | Up to 1,000 boosting rounds with early stopping after 50 rounds without validation improvement; maximum depth 5; learning rate 0.05; subsample 0.8; column subsample 0.8; L1 regularisation 0.1; L2 regularisation 1.0; log-loss evaluation metric; random seed 42 |

### *3.10 Graph Construction*

The graph is G = (V, E) with |V| = 21. The node set comprises nine index nodes (NIFTY 50 together with the eight sectoral indices), four macroeconomic nodes, seven country nodes and one global GDELT node, as set out in Table 3.8. Edges are grouped into six relation types totalling 56 unique undirected relationships, treated as bidirectional during message passing, as summarised in Table 3.9. The resulting structure is shown in Figure 3.3.

Edges encode domain-informed modelling assumptions intended to provide a relational inductive bias for message passing. The presence of an edge records an assumed association motivated by economic or geopolitical reasoning, and does not constitute evidence that one entity causes movement in another. The topology is fixed in code, applies unchanged across the entire sample period, and is not learned from the data.

**Table 3.8:** Graph node types.

| Node type | Members | Count |
|---|---|---|
| Index | NIFTY 50 and the eight sectoral indices | 9 |
| Macroeconomic | Gold, Oil, USD/INR, India VIX | 4 |
| Country | USA, India, China, Russia, Pakistan, Israel, Iran | 7 |
| Global | Aggregate GDELT signals | 1 |
| Total | | 21 |

**Table 3.9:** Graph edge groups, their rationale and their counts.

| Relation type | Rationale | Edges |
|---|---|---|
| Benchmark–Sector | Shared market-wide movement between NIFTY 50 and each sectoral index | 8 |
| Sector–Sector | Economically linked pairs such as Bank–Realty, Energy–Metal, Auto–Metal, FMCG–Pharma and IT–Pharma | 5 |
| Macro–Sector | Input-cost or revenue exposure, for example Oil–Energy, USD/INR–IT and VIX–Bank | 12 |
| Country–Sector | Assumed exposure to country-specific events, for example Iran–Energy, Russia–Energy and USA–IT | 12 |
| Country–Country | Bilateral relationships such as USA–China, India–Pakistan and Israel–Iran | 5 |
| Global–Other | Global GDELT activity linked to all countries, all macroeconomic nodes and three key index nodes | 14 |
| Total | | 56 |

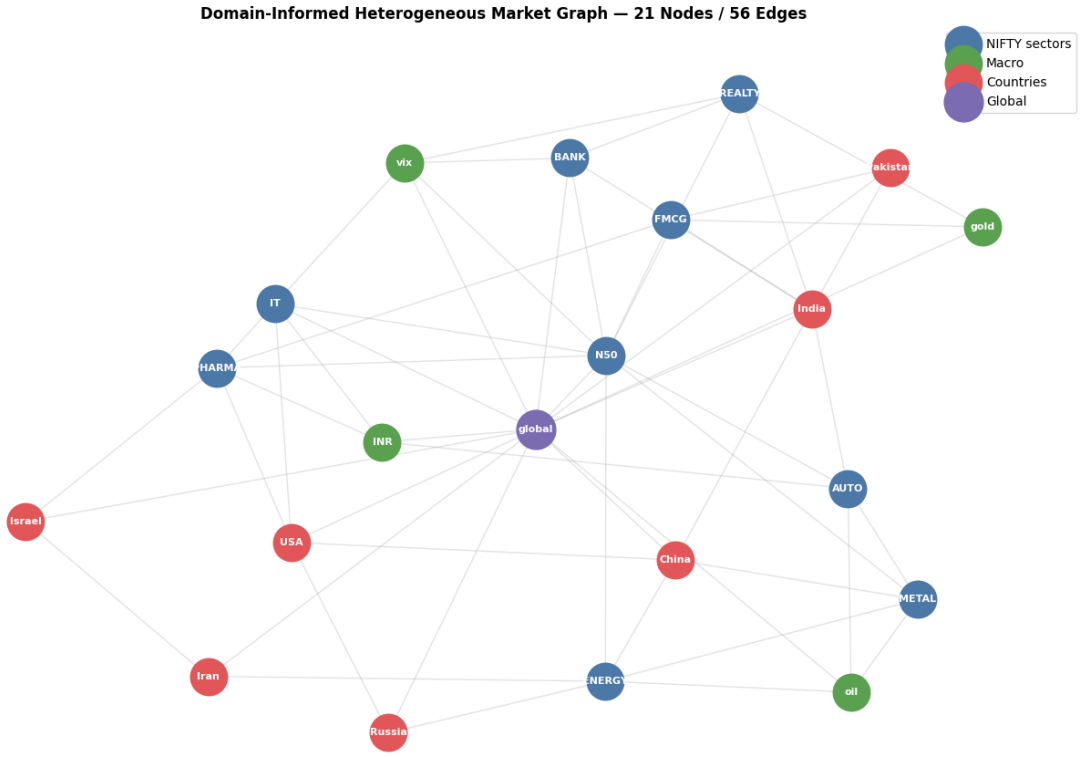



**Figure 3.3:** The 21-node, 56-edge domain-informed market graph, coloured by node type. Edges represent assumed relationships used as a modelling prior and not established causal links.

### *3.11 Node Representation*

Node features are assembled from 135 underlying engineered columns into a per-day tensor of shape 21 $\times$ 12, where 12 is the widest per-node feature set and narrower node types are zero-padded to match. Table 3.10 gives the allocation. Index nodes carry ten features: level, return, log return, 30-day moving average, volatility, momentum, relative strength, RSI, drawdown and price relative to the 30-day average. Macroeconomic nodes carry three: level, return and 30-day moving average. Country nodes carry three: conflict count, event count and average tone. The global node carries all twelve aggregate GDELT features.

**Table 3.10:** Node feature allocation within the 21 $\times$ 12 per-day tensor.

| **Node type** | **Features** | **Content** |
|---|---|---|
| Index | 10 | Level, return, log return, MA30, volatility, momentum, relative strength, RSI, drawdown, price/MA30 |
| Macroeconomic | 3 | Level, return, MA30 |
| Country | 3 | Conflict count, event count, average tone |
| Global | 12 | Twelve aggregate GDELT measures listed in Section 3.6 |

Zero padding is a pragmatic choice rather than a neutral one. After standardisation a padded slot holds the value zero, which coincides with the mean of a standardised feature, so padding is not distinguishable from an average observation. An architecture that maintained separate projection weights per node type would avoid this, and that possibility is noted in Section 6.6.

### *3.12 Edge Construction*

The adjacency matrix is built once from the 56 relationships and reused for every trading day, since the topology is fixed and only the node features vary over time. For the convolutional architectures the adjacency is symmetrically normalised with added self-loops, as given in Equation 3.2, and the result is computed once and shared across all forward passes.

$$\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2} \tag{3.2}$$

One implementation detail should be recorded for accuracy. The pre-normalised matrix $\hat{A}$ is the adjacency actually passed to the models, and the convolutional architectures apply the normalisation of Equation 3.2 again internally. The effective propagation operator is therefore a twice-normalised matrix with self-loops added twice. This does not invalidate the experiment, since the operator remains a fixed, symmetric, degree-weighted mixing matrix over the intended topology, but it does mean that the propagation weights differ slightly from a textbook GCN and it is reported here rather than passed over.

### *3.13 GNN Architectures*

All four models use two message-passing layers with a hidden dimension of 48, dropout of 0.20, and a linear output layer applied to the eight sectoral nodes only. The macroeconomic, country and global nodes participate fully in message passing but are never prediction targets, so they influence the outputs only through the representations they contribute to their neighbours. The GCN applies Equation 2.3 using the normalised adjacency. GraphSAGE concatenates each node's own representation with its aggregated neighbourhood before the linear transformation. GAT applies learned per-edge attention with four heads, masked to the graph topology. The HGT-style model adds a node-type embedding to the projected input, applies four-head attention with a learned per-relation bias added to the attention scores, and uses residual connections with layer normalisation. Configurations are summarised in Table 3.11.

**Table 3.11:** GNN configurations.

| Model | Layers | Hidden | Heads | Dropout | Learning rate |
|---|---|---|---|---|---|
| GCN | 2 | 48 | — | 0.20 | $2 \times 10^{-3}$ |
| GraphSAGE | 2 | 48 | — | 0.20 | $2 \times 10^{-3}$ |
| GAT | 2 | 48 | 4 | 0.20 | $1 \times 10^{-3}$ |
| HGT-style | 2 | 48 | 4 | 0.20 | $1 \times 10^{-3}$ |

### *3.14 GNN Training Procedure*

Training uses the Adam optimiser with weight decay $10^{-4}$, binary cross-entropy loss computed over the eight sectoral outputs of each graph, gradient clipping at a norm of 2.0, a batch size of 64 graphs and a learning-rate scheduler that halves the rate when validation loss stops improving.

The configuration executed to produce the reported results is an abbreviated one, with a maximum of 14 epochs and early-stopping patience of 4. A longer configuration, permitting 50 epochs with patience 10, exists in the codebase but was not the configuration run for these results, and no results from it are reported anywhere in this dissertation. This distinction materially affects the interpretation of Chapter 5 and is carried through to the discussion, the limitations and the future scope.

### *3.15 Leakage Prevention*

Four measures are applied. Splitting is chronological with no shuffling across partition boundaries. Imputation and scaling statistics are computed on the training partition only. Model selection uses the

validation partition, after which each graph model is re-initialised and retrained on the training and validation partitions combined for exactly the number of epochs identified as best, so that the validation data contributes to fitting only after its selection role is complete. The test partition is evaluated once, at the end, for each fitted model.

These measures address the leakage pathways that are under the control of the modelling pipeline. They do not establish that the experiment is free of all leakage. In particular, GDELT and macroeconomic values dated day t are assumed to be observable before the prediction point for day t+1, and exact publication timestamps were not available to verify this assumption. If a GDELT daily aggregate is finalised only after the Indian market close, part of that information would not have been available to a real-time user. This assumption is recorded in Section 6.4.

### *3.16 Evaluation Metrics*

Accuracy, F1 and ROC-AUC are reported on the held-out test partition. On this near-balanced task a random classifier corresponds to roughly 0.50 accuracy and exactly 0.50 AUC. Two further reference points recur in Chapter 5 and are not interchangeable: a constant-UP classifier, which always predicts the positive class and scores an accuracy equal to the UP prevalence in the data it is evaluated on, and a sector-wise majority-class classifier, which predicts whichever class is actually more frequent within each sector and can therefore differ from the constant-UP classifier whenever DOWN is the more frequent class in a given sector. Both are distinguished explicitly wherever they are used.

One methodological point must be stated clearly, because it affects every cross-pipeline comparison that follows. The two pipelines compute F1 differently. The classical pipeline computes macro F1, averaging the F1 score of both classes with equal weight. The graph pipeline computes positive-class binary F1, which considers only the UP class. These are different quantities, they respond differently to degenerate predictions, and they are not comparable. They are consequently labelled distinctly throughout this dissertation, and the consequences of the difference are analysed in Section 5.9. Accuracy and ROC-AUC are computed identically in both pipelines and are the valid basis for comparison between them. Statistical significance was not assessed anywhere in this study, because each reported configuration consisted of a single evaluation run on a single chronological split; no repeated-seed variance estimate, confidence interval or hypothesis test was computed, and no claim in this dissertation should be read as implying one was.

### *3.17 Explainability Method*

Interpretability is examined using gradient-based saliency applied to the best-performing graph model. Feature saliency is computed as the product of the absolute gradient of a selected sectoral output with respect to the input node-feature tensor and the absolute input values, averaged over 256 test-period samples. Edge saliency is computed as the absolute gradient of a selected sectoral output with respect to the dense adjacency matrix for a single representative test sample, with the two directions of each undirected edge averaged.

Two properties of this implementation constrain what can be concluded from it, and both are stated here rather than left implicit. First, the gradient is taken with respect to one selected output, the Auto sector, so every saliency value describes the sensitivity of that one output. Second, in the edge heatmap, edges that do not touch the target node are displayed at a fixed fraction of their computed saliency rather than being recomputed for each sector, so the rows of the heatmap are not independent per-sector explanations. Neither operation is a defect, but both limit interpretation, and Section 5.11 is written accordingly. This is not GNNExplainer and is not described as such anywhere in this dissertation.

### *3.18 Experimental Workflow*

The complete workflow runs from data reconstruction and integration, through preprocessing and feature construction, target creation and chronological splitting, into the two parallel modelling tracks, and finally to evaluation on the respective held-out test partitions, which overlap chronologically but differ slightly in row count and ending date, followed by gradient-based analysis of the best graph model. Each stage is executed once, in order, with fixed random seeds, and no stage consults the test partition before the final evaluation.

# CHAPTER 4
# IMPLEMENTATION

## *4.1 Software Environment*

The classical pipeline uses scikit-learn for the Random Forest implementation, median imputation and metric computation, together with the XGBoost library [5] for the gradient-boosted models. The graph pipeline uses PyTorch 2.10.0. PyTorch Geometric is deliberately not used: all four architectures are implemented directly as tensor operations over a dense adjacency matrix, so that the notebook executes in an environment where that library is unavailable. This choice has a direct consequence for interpretability, because the GNNExplainer implementation distributed with PyTorch Geometric is correspondingly unavailable, which is one reason gradient-based saliency is used instead. NetworkX is used for the graph visualisation in Figure 3.3 and Matplotlib for all remaining figures. Table 4.1 summarises the environment.

**Table 4.1:** Software environment.

| **Component** | **Version or setting** |
|---|---|
| Python | 3.x |
| PyTorch | 2.10.0 (CPU build) |
| PyTorch Geometric | Not used |
| scikit-learn | Random Forest, median imputation, metrics |
| XGBoost | Gradient-boosted trees with early stopping |
| pandas / NumPy | Data reconstruction, alignment and feature construction |
| NetworkX / Matplotlib | Graph visualisation and all result figures |

## *4.2 Hardware Environment*

All training was performed on CPU; the device check in the graph notebook reports that no CUDA device was available, and the PyTorch build is a CPU build. This was practical because the graph is small, the per-day tensor has only 252 entries and the executed training run was abbreviated. The total training time across all four graph models was under seven seconds, as recorded in Table 5.8. Hardware was therefore not a binding constraint on the results reported here, and any future extension to longer training and systematic hyperparameter search would remain feasible on modest hardware.

**Table 4.2:** Hardware environment.

| **Component** | **Setting** |
|---|---|
| Processor | CPU execution only |
| GPU | None used; no CUDA device detected at run time |
| Total GNN training time | 6.72 s across the four architectures |

## *4.3 Dataset Processing*

Both notebooks begin by attempting to load a prebuilt merged dataset and, on finding it absent, reconstruct an equivalent modelling table from the component files. The classical notebook reports this fallback explicitly at run time, which is how the reconstruction path was identified during the audit of the executed outputs. Date columns are located by regular-expression matching because the source market files carry

a two-row header. Series are aligned on date, the targets are constructed using Equation 3.1, and the final row is dropped because its next-day outcome is unobservable.

### *4.4 Classical ML Implementation*

For each combination of feature configuration and sector, both models are fitted on the training partition after median imputation fitted on that partition. XGBoost monitors the validation partition for early stopping; Random Forest is fitted directly. Each fitted model is then evaluated once on the test partition, producing accuracy, macro F1 and ROC-AUC. Metrics are collected per sector and averaged across the eight sectors to give the configuration-level figures reported in Section 5.3. Feature importances are retained for each fitted model, although they are not used for the comparative conclusions of this study.

### *4.5 GDELT Ablation Implementation*

The ablation is implemented by constructing the two feature matrices described in Section 3.5 and running an identical training and evaluation procedure over each. Everything else is held constant, including the chronological split, the hyperparameters, the imputation strategy and the random seed, so the difference between the E1 and E2 results is attributable to the presence or absence of the 33 GDELT features and not to any other change in the pipeline.

### *4.6 Graph Construction Implementation*

Node indices are assigned once, in a fixed order, and the 56 relationships are expanded into a symmetric dense adjacency matrix. Duplicate pairs arising across relation groups are removed before expansion, so the reported count of 56 is a count of unique undirected relationships rather than of listed pairs. The normalised adjacency of Equation 3.2 is precomputed and reused across all forward passes, and a parallel relation-identifier matrix is retained so that the HGT-style model can condition its attention on relation type. Per-day node features are assembled into a $21 \times 12$ tensor with zero padding for node types carrying fewer features, and days are stacked into mini-batches of 64 graphs.

### *4.7 GNN Implementation*

Each architecture is implemented as a compact PyTorch module operating on a batched tensor of shape [batch, 21, 12] together with the shared adjacency. Message passing is expressed as batched matrix multiplication for the convolutional models and as masked multi-head attention for GAT and the HGT-style model, with the mask derived from the adjacency so that attention is computed only over existing edges. The output layer is indexed to node positions one through eight before the loss is computed, so that only sectoral nodes contribute to the objective while the remaining nodes still influence predictions through message passing. The HGT-style model additionally receives node-type and relation-type tensors so that its attention can condition on them, and retains the adjacency magnitude inside the attention score so that a gradient with respect to the adjacency is defined, which is what makes the edge saliency of Section 3.17 computable.

### *4.8 Training and Validation Procedure*

Training follows the two-stage procedure described in Section 3.15. Each model is first trained on the training partition with validation loss monitored after every epoch, both to drive the learning-rate scheduler and to identify the epoch at which validation loss is lowest. The model is then re-initialised from scratch and retrained on the combined training and validation set of 1,513 samples for exactly that number of epochs. Test evaluation is performed once, afterwards, and its outputs are not used to adjust any part of the pipeline.

### *4.9 Explainability Implementation*

Gradient-based saliency is applied to the model with the highest test ROC-AUC. Feature saliency requires gradients with respect to the input tensor, computed on a batch of 256 test samples and averaged. Edge saliency requires gradients with respect to the adjacency matrix and is computed from the first test sample with the target fixed to a single sector, which is the constraint discussed in Section 3.17 and revisited in Section 5.11.

### *4.10 Reproducibility*

Random seeds are fixed at 42 for the data handling, model initialisation and training loops in both pipelines. The graph topology is defined statically in code rather than learned, so it reproduces exactly. Hyperparameters are fixed in the configuration cell rather than searched, and the abbreviated training configuration is controlled by a single flag whose value at execution time was recorded during the audit.

The main reproducibility caveat concerns the dataset itself. Because the modelling table is reconstructed from component files at run time rather than loaded from a fixed merged file, reproducing these results requires the same component files in the same form. Anyone repeating the work should first confirm the reconstructed row count of 2,086 for the classical pipeline, the feature-group counts in Table 3.4, and the partition sizes in Table 3.5 and Table 3.6 before comparing any model metric, since a difference in any of these would indicate that a different dataset has been assembled.

# CHAPTER 5
# RESULTS AND DISCUSSION

## *5.1 Dataset Characteristics*

The reconstructed classical modelling table contains 2,086 trading days spanning 1 January 2018 to 19 June 2026 across 114 raw columns, from which 63 sector-technical, 21 country-level GDELT and 12 global GDELT features are drawn. The chronological split gives the classical pipeline 1,234 training days, 365 validation days and 487 test days. The graph pipeline is slightly smaller at 1,161, 352 and 470 respectively, for the join and completeness reasons set out in Section 3.3; its test period ends two trading days earlier, on 17 June 2026, because the inner join requires every series to be present.

The test partition therefore covers just under two years of trading, which is long enough to span more than one market phase and short enough that a single unusual episode could influence the aggregate metrics. This is relevant to the interpretation of small differences between configurations and is returned to in Section 5.14.

## *5.2 Class Distribution*

Table 5.1 shows the proportion of UP days per sector over the full reconstructed classical dataset. All eight sectors lie between 51.0% and 53.7%, a mild majority-class tilt consistent with the general upward drift of equity indices over the period.

**Table 5.1:** Class balance per sector over the full classical dataset (percentage of UP days).

| **Sector** | **UP (%)** | **DOWN (%)** |
|---|---|---|
| Auto | 51.4 | 48.6 |
| Bank | 53.2 | 46.8 |
| Energy | 53.1 | 46.9 |
| FMCG | 51.0 | 49.0 |
| IT | 52.3 | 47.7 |
| Metal | 53.7 | 46.3 |
| Pharma | 51.4 | 48.6 |
| Realty | 52.3 | 47.7 |

Table 5.2 gives the corresponding positive-class rates by partition for the graph pipeline, which are the figures that matter for interpreting the graph results. The mean test prevalence across the eight sectors is 0.5169, and this value reappears in Section 5.8 in a way that is not coincidental.

**Table 5.2:** Positive-class rate by partition in the graph pipeline.

| Sector | Train | Validation | Test |
|---|---|---|---|
| Auto | 0.508 | 0.582 | 0.509 |
| Bank | 0.534 | 0.534 | 0.532 |
| Energy | 0.530 | 0.577 | 0.526 |
| FMCG | 0.518 | 0.531 | 0.485 |
| IT | 0.545 | 0.489 | 0.498 |
| Metal | 0.528 | 0.557 | 0.564 |
| Pharma | 0.497 | 0.557 | 0.534 |
| Realty | 0.523 | 0.591 | 0.487 |

Two observations follow. The class balance shifts between partitions, most visibly for Realty, which moves from 0.523 in training to 0.591 in validation and 0.487 in test, and for Auto, which moves from 0.508 to 0.582 and back to 0.509. A model that learns a partition-specific class prior rather than a genuine decision rule will therefore be penalised, and validation-based model selection will be correspondingly noisy. More importantly, a constant-UP predictor would score roughly 52% accuracy on this data without learning anything, which is close to every accuracy figure reported in the remainder of this chapter. Accuracy alone therefore cannot distinguish a working model from a degenerate one, which is why AUC and the prediction distributions are examined alongside it throughout.

### *5.3 Classical Machine Learning Results*

Table 5.3 reports mean test performance across the eight sectors for all four classical configurations. The mean accuracy, macro F1 and ROC-AUC for the top-ranked configuration are taken directly from the notebook summary output; the aggregate figures for the remaining three configurations were recomputed from the per-sector outputs, because the notebook printed aggregate F1 and AUC only for its top-ranked configuration. Both sources agree to within rounding for the configuration where they overlap.

**Table 5.3:** Classical ML mean test performance across the eight sectors. F1 is macro-averaged.

| Model | Configuration | Accuracy | Macro F1 | ROC-AUC |
|---|---|---|---|---|
| XGBoost | E2 (Technical Only) | 51.67% | 44.23% | 0.5134 |
| Random Forest | E1 (Technical + GDELT) | 50.77% | 44.41% | 0.5315 |
| Random Forest | E2 (Technical Only) | 50.62% | 42.93% | 0.5310 |
| XGBoost | E1 (Technical + GDELT) | 50.03% | 39.56% | 0.5060 |
| Random baseline | — | ≈50% | — | 0.5000 |

The random baseline in Table 5.3 is a coin-flip reference (0.50 accuracy, 0.50 AUC) and is distinct from a majority-class baseline, which would instead score the prevalence of the more frequent class. A majority-class baseline could not be computed for the classical test partitions specifically, because the executed notebook printed the class balance only for the full 2,086-row dataset (Table 5.1, 51.0% to 53.7% UP across sectors) and not for the 487-day test partition in isolation; substituting the full-sample figure for a test-period majority baseline would risk overstating or understating it, so it is left unreported here rather than approximated. This gap is listed as a limitation in Section 6.4, and reconstructing it is listed as future work in Section 6.6.

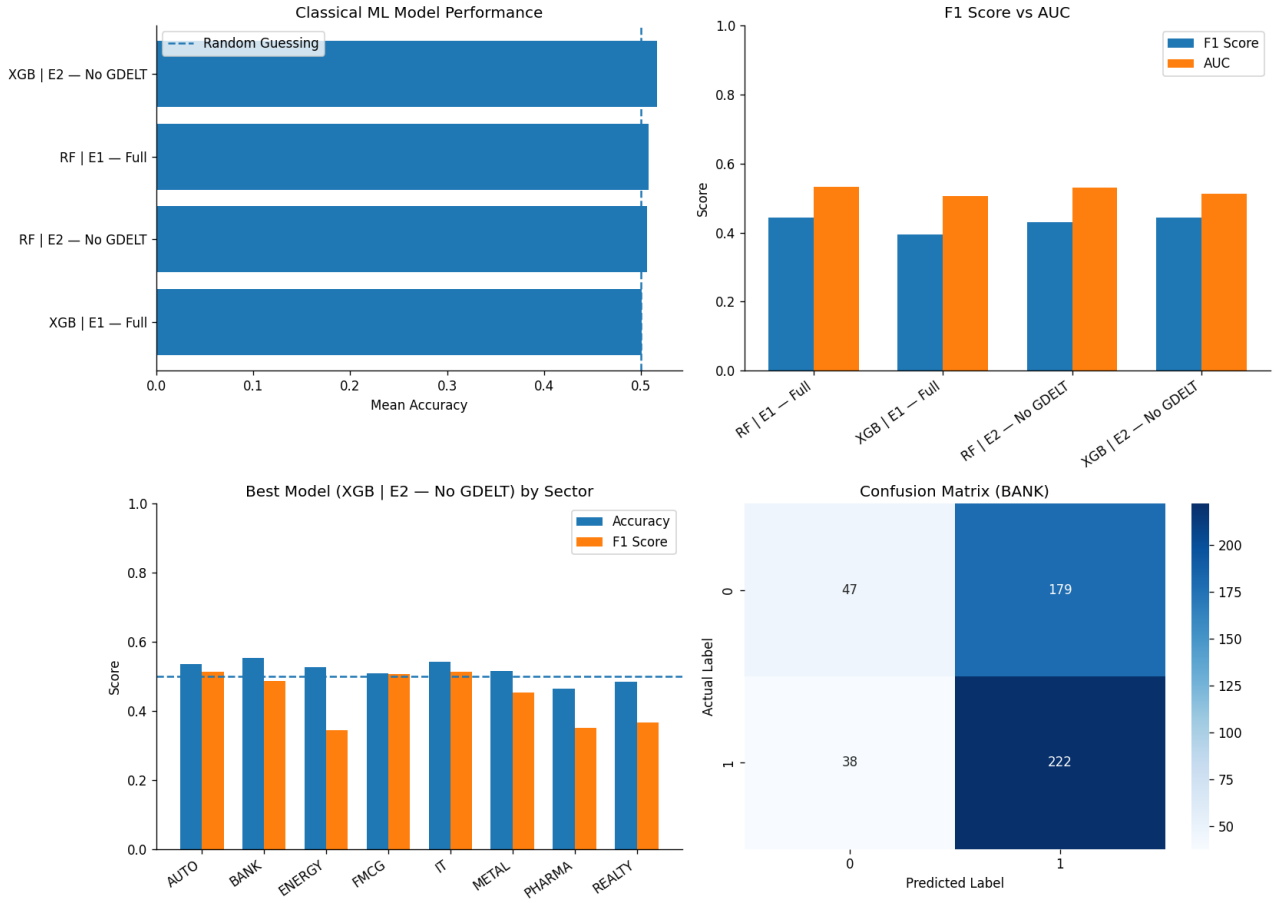



**Figure 5.1:** Classical ML results. Top left: mean accuracy by configuration against the random-guessing line. Top right: mean F1 and AUC by configuration. Bottom left: per-sector accuracy and macro F1 for XGBoost without GDELT features. Bottom right: confusion matrix for that configuration on the Bank sector.

The spread between the best and worst configuration is 1.64 percentage points of accuracy, and every configuration sits within roughly two points of the random baseline. The strongest result, 51.67% accuracy at 0.5134 AUC, is above chance but by a margin small enough that no claim of practical predictive skill is warranted, particularly given that a constant-UP predictor would score around 52% on this data. The honest summary is that the evaluated XGBoost configuration produced the strongest average accuracy among the tested classical models, while its advantage over a trivial baseline remained negligible.

An inconsistency in ranking deserves comment rather than concealment. Ordered by accuracy, XGBoost without GDELT features is best. Ordered by ROC-AUC, Random Forest with GDELT features is best at 0.5315. The two orderings disagree because accuracy depends on a fixed 0.5 decision threshold whereas AUC assesses the ranking of predicted probabilities across all thresholds; a model can rank observations slightly better while placing its threshold slightly worse. With differences of this magnitude, neither ordering should be treated as establishing a genuinely superior configuration, and both are reported in Table 5.3 so that the disagreement is visible to the reader.

### *5.4 GDELT Ablation Results*

Table 5.4 isolates the effect of the 33 GDELT features by pairing each model's two configurations. A positive value indicates that removing the GDELT features improved the metric.

**Table 5.4:** Effect of removing GDELT features (E1 → E2). Positive values favour the configuration without GDELT.

| Model | Δ Accuracy | Δ Macro F1 | Δ ROC-AUC |
|---|---|---|---|
| XGBoost | +1.64 pts | +4.67 pts | +0.0074 |
| Random Forest | -0.15 pts | -1.48 pts | -0.0005 |

The two algorithms respond differently. Removing the GDELT features improved XGBoost on all three metrics, most noticeably on macro F1. For Random Forest the changes are negligible in every case, with an AUC difference of 0.0005 that is far too small to carry meaning given single-run point estimates.

The answer to RQ1 is therefore that the evaluated GDELT-derived features did not improve next-day directional prediction under the tested classical configurations. The evidence is mixed in direction but consistent in magnitude: where a difference appears it is small, and where it is largest it favours the configuration without GDELT. One plausible explanation is that 33 additional noisy features increase the dimensionality of an already difficult problem without carrying proportionate signal, which would be expected to affect a sequentially fitted boosting model more than an averaged bagging ensemble, since boosting can allocate splits to spurious structure in the residuals. That interpretation is consistent with the observed pattern but is not established by this experiment. A further consideration, noted in Section 3.5, is that the ablation was performed on top of sector-technical features alone, because no dedicated macroeconomic columns were present in the reconstruction; the result may not generalise to a configuration with fuller macroeconomic coverage.

### *5.5 Sector-Wise Classical Results*

Table 5.5 breaks the top-ranked configuration down by sector, and Table 5.6 gives per-sector accuracy for all four configurations so that the sector-level variation can be seen across models rather than for one configuration alone.

**Table 5.5:** Per-sector test performance, XGBoost E2 (Technical Only).

| Sector | Accuracy | Macro F1 | ROC-AUC |
|---|---|---|---|
| Bank | 0.553 | 0.487 | 0.556 |
| IT | 0.543 | 0.514 | 0.494 |
| Auto | 0.535 | 0.514 | 0.542 |
| Energy | 0.527 | 0.345 | 0.526 |
| Metal | 0.516 | 0.453 | 0.466 |
| FMCG | 0.508 | 0.508 | 0.505 |
| Realty | 0.486 | 0.368 | 0.505 |
| Pharma | 0.465 | 0.351 | 0.514 |

**Table 5.6:** Per-sector test accuracy for all four classical configurations.

| Sector | RF E1 | XGB E1 | RF E2 | XGB E2 |
|---|---|---|---|---|
| Auto | 0.514 | 0.477 | 0.484 | 0.535 |
| Bank | 0.469 | 0.521 | 0.479 | 0.553 |
| Energy | 0.541 | 0.527 | 0.547 | 0.527 |
| FMCG | 0.516 | 0.494 | 0.529 | 0.508 |
| IT | 0.508 | 0.479 | 0.506 | 0.543 |
| Metal | 0.539 | 0.560 | 0.510 | 0.516 |
| Pharma | 0.475 | 0.465 | 0.467 | 0.465 |
| Realty | 0.498 | 0.479 | 0.527 | 0.486 |

Performance varies by roughly nine accuracy points across sectors for the top-ranked configuration, from Bank at 0.553 to Pharma at 0.465. Two sectors fall below 0.50 accuracy while retaining AUC above 0.50, Pharma at 0.514 and Realty at 0.505, which indicates a threshold effect rather than inverted predictions: the ranking of predicted probabilities carries slightly more information than the thresholded labels do, and a tuned decision threshold might recover part of it.

The sector ordering is not stable across configurations. Metal is the strongest sector for XGBoost with GDELT at 0.560 but only mid-range for the top-ranked configuration at 0.516; Bank is worst for Random Forest with GDELT at 0.469 and best for XGBoost without GDELT at 0.553. Sector-level patterns at these effect sizes are therefore not reliable, and no economic interpretation is offered for them.

Energy is worth noting separately because it is the sector most directly associated with the geopolitical entities represented in the graph, being linked to Russia, Iran, China and oil. Its accuracy of 0.527 is mid-range and its macro F1 of 0.345 is among the lowest in Table 5.5, so there is no indication here that geopolitically exposed sectors were easier to predict. This is consistent with the ablation result of Section 5.4.

### *5.6 GNN Results*

Table 5.7 reports the four graph architectures on the held-out test period. The F1 column contains positive-class binary F1 and is not comparable with the macro F1 of Table 5.3; Section 5.9 examines the consequences of that difference in detail.

**Table 5.7:** GNN test performance. F1 is positive-class binary F1.

| Model | Accuracy | Binary F1 | ROC-AUC |
|---|---|---|---|
| HGT-style | 0.5168 | 0.6814 | 0.5018 |
| GAT | 0.5168 | 0.6814 | 0.4936 |
| GCN | 0.5168 | 0.6814 | 0.4858 |
| GraphSAGE | 0.5168 | 0.6814 | 0.4855 |
| Random baseline | $\approx$0.50 | — | 0.5000 |
| Constant-UP baseline (test) | 0.5169 | — | not applicable |
| Sector-wise majority-class baseline (test) | 0.5242 | — | not applicable |

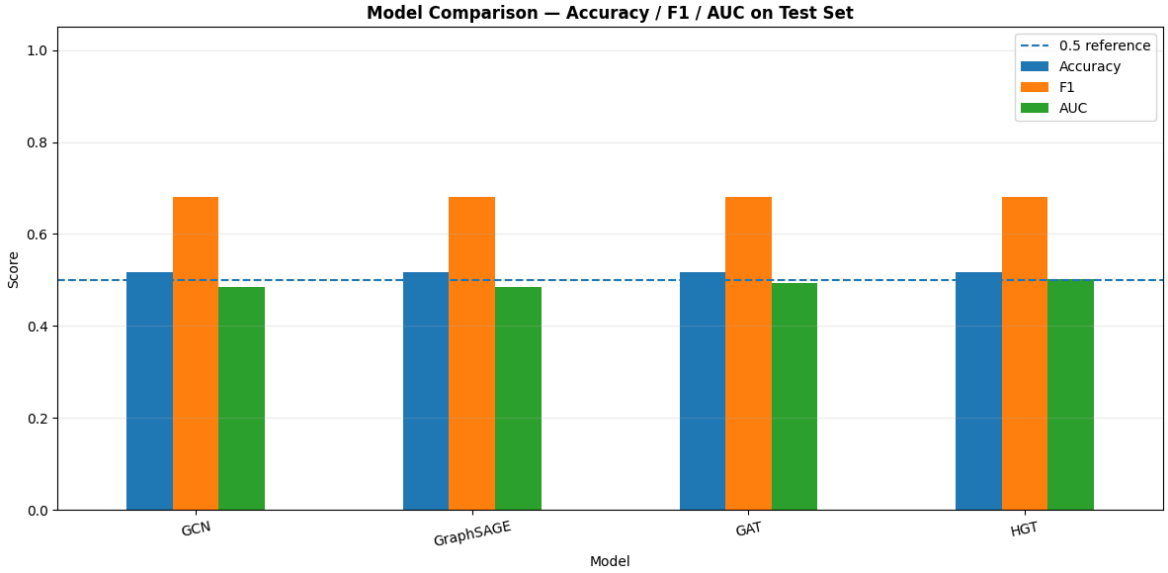



**Figure 5.2:** Aggregate accuracy, F1 and ROC-AUC for the four graph architectures against the 0.5 reference line.

Three distinct reference points appear in Table 5.7, and they are not interchangeable. The random baseline (0.50 accuracy, 0.50 AUC) is a coin-flip reference. The constant-UP baseline, 0.5169, is the mean test-period UP prevalence across the eight sectors from Table 5.2; it is what any classifier that always predicts UP would score, and it is the figure the graph models' own accuracy should be read against, since that is exactly what they did. The sector-wise majority-class baseline, 0.5242, is different and is not simply a restatement of the same idea: it is obtained by predicting, for each sector, whichever class was actually more frequent in that sector's 470 test observations, then averaging the resulting accuracy across the eight sectors. Because DOWN was in fact the more frequent class in three sectors during the test period (FMCG, IT and Realty, from the confusion-matrix counts in Table 5.9), always predicting UP is not the optimal constant strategy for this test period, and the sector-wise majority baseline is consequently higher than the constant-UP baseline. Neither figure is a majority-class baseline for the classical pipeline, which Section 5.3 explains could not be reconstructed from the available output; both apply only to the graph test partition, where the full per-sector confusion-matrix counts needed to compute them are available. ROC-AUC is not applicable to either baseline, since a constant classifier produces no ranking over cases; this is unrelated to the AUC reported for the graph models themselves, discussed next.

Measured against the constant-UP baseline of 0.5169, every graph model's accuracy of 0.5168 is effectively equal to it at the reported precision, which is unsurprising given what Section 5.8 shows about these predictions: the models did not merely approach the constant-UP baseline, they achieved it by degenerating to exactly that strategy. Measured instead against the higher sector-wise majority-class baseline of 0.5242, the graph models sit slightly below it. Either comparison is a materially more informative reading than measuring 0.5168 against a 0.50 random baseline, and both are reported here rather than only the more flattering one. This is a descriptive comparison of point values, not a statistical test; no significance test or confidence interval was computed, so "effectively equal" is not a claim of statistical indistinguishability. Although each graph model's thresholded predictions are constant, its underlying predicted probabilities vary across observations, which is what allows a real ROC-AUC to be computed for it in the first place. The graph-model AUC values in Table 5.7 are therefore genuine computed statistics describing probability ranking, not artefacts of the constant labels, and are interpreted cautiously below precisely because that ranking is weak, not because the statistic is undefined.

Three observations follow immediately. Accuracy and F1 are identical to four decimal places across

all four models, which is not what four independently initialised and independently trained architectures would normally produce. Only the HGT-style model exceeds 0.50 AUC, and only by 0.0018. GCN and GraphSAGE fall below the random line.

The answer to RQ2 is negative. No graph architecture outperformed the strongest classical configuration on the metrics that permit comparison: the best graph AUC of 0.5018 is below the best classical AUC of 0.5315 and below the 0.5134 of the classical configuration that ranked first on accuracy. Stated carefully, the HGT-style model obtained the highest ROC-AUC among the evaluated graph configurations, but the resulting value remained essentially at the random-classification level.

The AUC ordering across the four models spans 0.0163, from 0.4855 to 0.5018. Given single-run point estimates and the absence of significance testing, this ordering should not be read as evidence that type-aware attention is better suited to the task; no formal statistical test was performed to support a claim of significant separation between the models. It is more safely described as four models whose AUC values sit close to chance and to one another, against the degenerate prediction behaviour documented in Section 5.8.

### *5.7 GNN Training Behaviour*

Table 5.8 records the selected epoch, the best validation loss and the training time for each architecture, and Figure 5.3 shows the corresponding loss curves.

**Table 5.8:** GNN training behaviour under the abbreviated configuration.

| **Model** | **Best validation BCE** | **Training time (s)** |
|---|---|---|
| GCN | 0.6875 | 2.33 |
| GraphSAGE | 0.6876 | 0.63 |
| GAT | 0.6887 | 1.49 |
| HGT-style | 0.6950 | 2.28 |

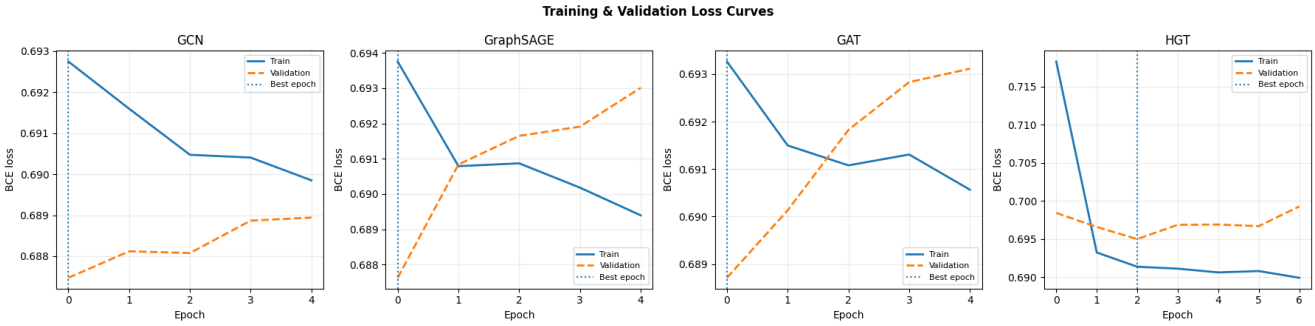



**Figure 5.3:** Training and validation binary cross-entropy for the four architectures, with the selected best epoch marked by a vertical line.

For GCN, GraphSAGE and GAT the pattern is the same: validation loss reaches its minimum at epoch 1 and rises thereafter while training loss declines slowly. The corresponding best validation losses of 0.6875, 0.6876 and 0.6887 should be compared with $\ln 2 \approx 0.6931$, which is the loss of a model that assigns equal probability to both classes. The models therefore barely move away from an uninformative solution before validation performance begins to degrade. The HGT-style model behaves somewhat differently, starting from a higher loss and improving over three epochs to 0.6950, which is still close to the same level and in fact slightly above it.

Because early stopping selected epoch 1 for three of the four models, the subsequent refit on the combined training and validation set ran for a single epoch. What Table 5.7 reports is therefore the

test performance of models that received one or three epochs of gradient descent under the abbreviated configuration described in Section 3.14. These figures characterise that configuration. They are not an upper bound on what these architectures could achieve with longer training and systematic hyperparameter search, and no claim to that effect is made anywhere in this dissertation.

### *5.8 Confusion Matrix Analysis*

The identical accuracy and F1 values across four different architectures prompted direct inspection of the predictions rather than acceptance of the aggregate metrics. Figure 5.4 shows the per-sector test confusion matrices for the HGT-style model and Table 5.9 gives the same counts in numerical form.

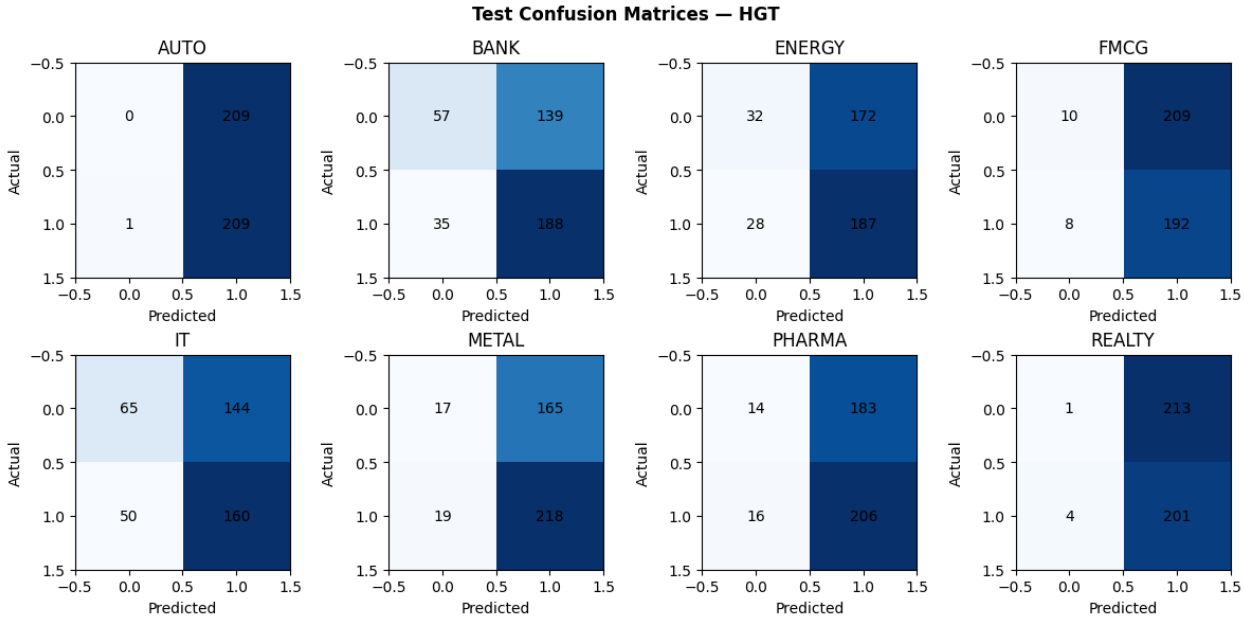



**Figure 5.4:** Per-sector test confusion matrices for the HGT-style model. The predicted-DOWN column is empty in every sector.

**Table 5.9:** Test confusion-matrix counts for the HGT-style model, per sector (470 test days).

| Sector | Actual DOWN → pred UP | Actual UP → pred UP | Predicted DOWN | Test UP rate |
|---|---|---|---|---|
| Auto | 231 | 239 | 0 | 0.509 |
| Bank | 220 | 250 | 0 | 0.532 |
| Energy | 223 | 247 | 0 | 0.526 |
| FMCG | 242 | 228 | 0 | 0.485 |
| IT | 236 | 234 | 0 | 0.498 |
| Metal | 205 | 265 | 0 | 0.564 |
| Pharma | 219 | 251 | 0 | 0.534 |
| Realty | 241 | 229 | 0 | 0.487 |

Every test observation in every sector is assigned to the UP class. The predicted-DOWN column is empty throughout, and the same behaviour holds for GCN, GraphSAGE and GAT. This explains the identical accuracy and F1 figures directly: at a 0.5 threshold all four models produce exactly the same labels, namely all-positive, so any metric computed from labels alone must agree across them. The models differ only in the underlying probability scores, which is precisely why AUC differs while accuracy and F1 do not. The arithmetic confirms it: the mean test prevalence of 0.5169 from Table 5.2 matches

the reported accuracy of 0.5168 to within rounding, which is what an all-positive predictor must achieve by construction.

The answer to RQ3 is therefore negative. Under the training configuration used, none of the four architectures learned a discriminative decision boundary on the held-out test period; all four GNN models produced constant-UP predictions on the held-out test observations. Several factors plausibly contribute, and the experiment as designed cannot separate them. The models carry on the order of tens of thousands of parameters but were fitted on 1,513 training graphs for one to three epochs, which supplies very little gradient signal relative to the parameter count. The task itself has a low signal-to-noise ratio, so the loss surface near the uninformative solution is close to flat. The mild class imbalance documented in Table 5.2 makes constant-UP prediction a locally attractive solution that a weakly trained model can settle into, particularly when the loss is binary cross-entropy averaged over eight outputs of differing prevalence. Distinguishing these explanations would require the longer training runs described in Section 6.6, which is the first item of proposed future work for exactly this reason.

### *5.9 The F1 Metric Discrepancy*

The graph models report F1 of 0.6814 while the best classical configuration reports 44.23%. Taken at face value this would suggest that the graph models are substantially better, which the rest of the evidence contradicts. The explanation is that the two pipelines compute different metrics, and the discrepancy is worth setting out in full because it is the kind of error that is easy to reproduce in a comparative study.

With all predictions positive, recall on the UP class is 1 by construction and precision equals the class prevalence p, so positive-class F1 reduces to the expression in Equation 5.1.

$$F_1^{\text{binary}} = \frac{2p}{1+p} \tag{5.1}$$

Substituting the observed mean test prevalence p = 0.5168 gives 0.6814, which matches the reported value exactly. This confirms both that the metric is positive-class binary F1 and that the predictions are entirely positive; the two facts are recoverable from the metric alone.

Computed the way the classical pipeline computes it, as a macro F1 averaged per sector, the constant-UP predictions score approximately 0.3405. This figure is derived from the reported per-sector confusion matrices in Table 5.9 rather than recomputed from stored per-sample predictions, since those were not retained: for each sector, the DOWN-class F1 is 0 because recall on DOWN is 0 under a constant-UP rule, the UP-class F1 follows from that sector's own UP prevalence, and the sector's macro F1 is the average of the two; the eight per-sector macro-F1 values are then averaged, exactly matching how the classical macro-F1 figures in Table 5.3 were aggregated. Table 5.10 places the two definitions side by side. On a common metric the graph models score approximately 0.3405 against 0.4423 for the best classical configuration, which reverses the apparent ordering entirely. The high graph F1 is an artefact of the metric definition combined with constant-UP predictions, and is not evidence of performance.

**Table 5.10:** The same graph predictions scored under both F1 definitions, with the best classical configuration for reference.

| **Configuration** | **Positive-class F1** | **Macro F1** |
|---|---|---|
| Graph models (all four, constant-UP predictions) | 0.6814 | ≈0.3405 |
| XGBoost E2 (Technical Only), classical | not computed | 0.4423 |

Accuracy and ROC-AUC are computed identically in both pipelines and remain valid for comparison. Every cross-pipeline conclusion drawn in this dissertation rests on those two metrics, and the derived

macro F1 of approximately 0.3405 is reported alongside them rather than being used as an independent result, since it is derived from the reported per-sector confusion matrices rather than recomputed from stored per-sample predictions.

### *5.10 Per-Sector GNN Results*

Figure 5.5 shows per-sector test AUC for all four architectures. Exact per-sector AUC values were not written to the notebook output, so they are presented graphically and the discussion below is limited to what can be read reliably from the chart.

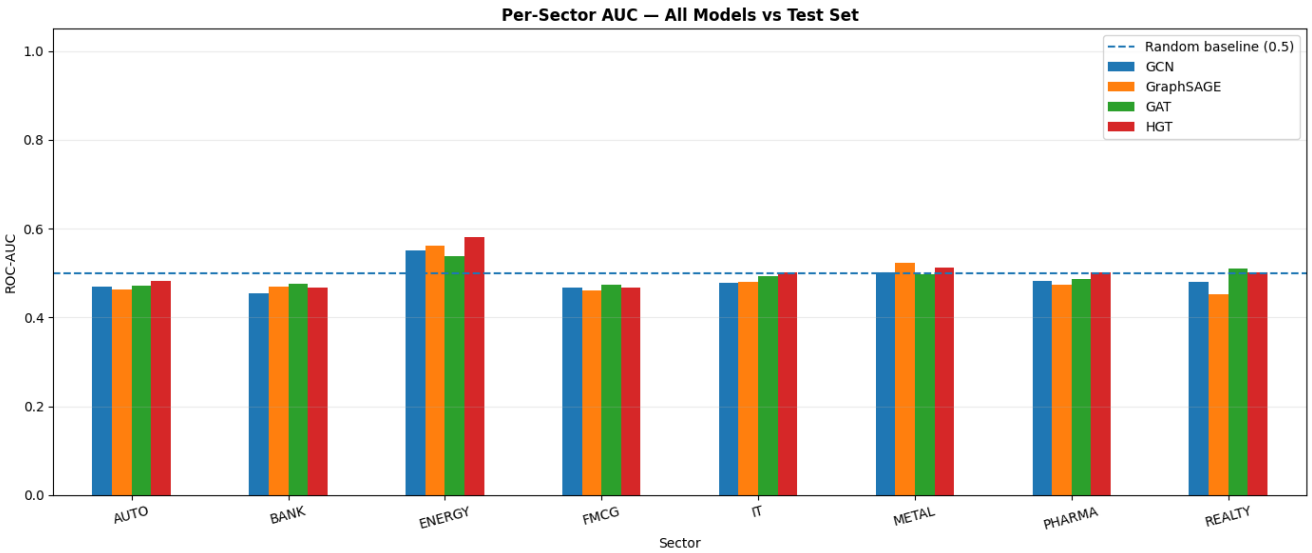



**Figure 5.5:** Per-sector test ROC-AUC for the four graph architectures against the 0.5 random baseline.

Energy is the one sector where all four models sit consistently above 0.50, in the region of 0.54 to 0.58. Most other sectors fall at or slightly below the line, with Metal and Realty marginally above it for some architectures. Because every model predicts UP everywhere, this cannot indicate a working classifier for Energy; it indicates only that the predicted probabilities rank Energy observations somewhat better than chance, while the thresholded labels remain constant.

It is worth noting that Energy was not a strong sector for the classical models, where its macro F1 was among the lowest in Table 5.5. The two pipelines therefore disagree about which sectors are relatively tractable, which is a further reason to treat sector-level patterns at these effect sizes as tentative rather than as evidence of differential predictability.

### *5.11 Explainability Results*

Figure 5.6 shows gradient-based feature saliency for the HGT-style model. Each panel corresponds to one sectoral node and shows how strongly the selected output responds to that node's ten technical features.

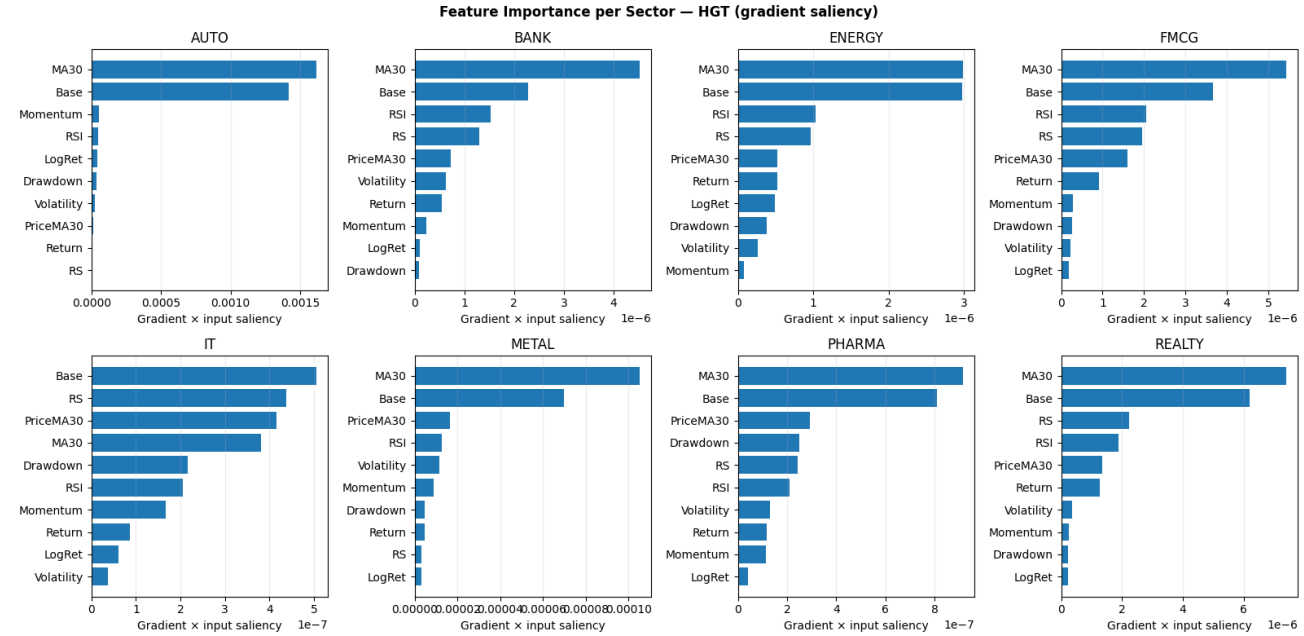



**Figure 5.6:** Gradient-based feature saliency for the HGT-style model, computed as |gradient| $\times$ |input| and averaged over 256 test samples. The gradient is taken with respect to a single sectoral output, so each panel shows how that output responds to the features of the corresponding node.

The 30-day moving average and the raw price level dominate in nearly every panel, well ahead of momentum, RSI and volatility. One reading is that the model responds most to slow-moving, strongly autocorrelated inputs rather than to shorter-horizon indicators, which would be consistent with a model that has settled into a near-constant output; a nearly constant function has its largest sensitivities where the inputs themselves are largest in magnitude. This is an interpretation rather than a demonstrated mechanism, and it should be weighed against the fact that saliency computed on a degenerate model describes the sensitivity of a function that barely varies. The absolute saliency magnitudes, on the order of $10^{-6}$ to $10^{-4}$, are themselves consistent with that reading.

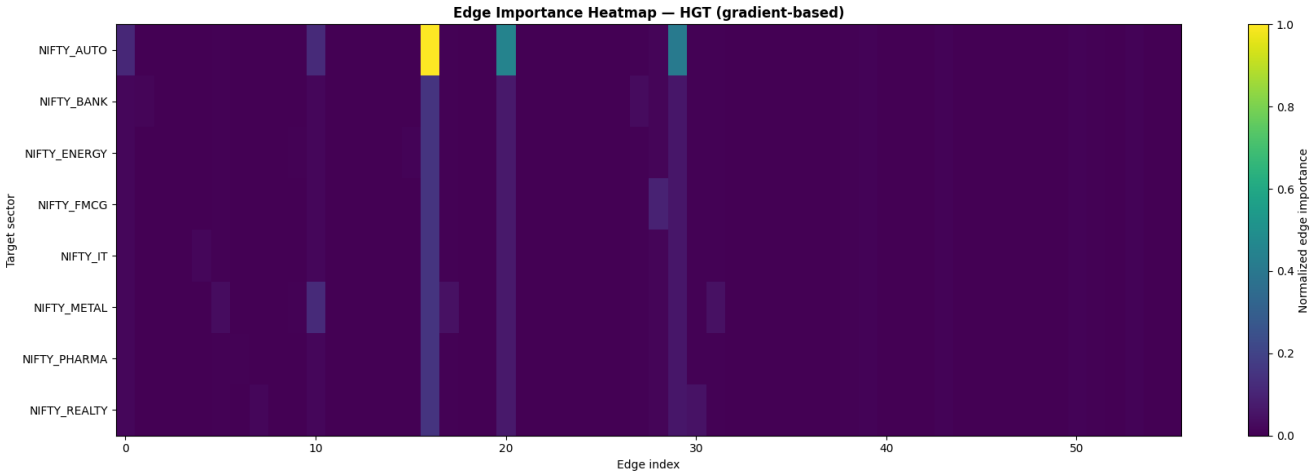



**Figure 5.7:** Gradient-based edge-importance heatmap for the HGT-style model, computed with the target output fixed to a single sector.

Figure 5.7 shows the edge saliency. A small number of edges register elevated importance while most of the 56 show very little. As stated in Section 3.17, this computation used a single fixed target sector, so the values describe a sensitivity profile for that one output; the remaining rows of the heatmap are derived from the same gradient rather than from independent per-sector computations. For that reason no claim

is made here about which specific country-to-sector relationships matter for which sectors. Establishing that would require an independent saliency pass per sector applied to a model that is not degenerate, which is listed in Section 6.6.

### *5.12 Classical ML versus GNN*

Table 5.11 places the two pipelines side by side on the metrics that are computed identically in both.

This comparison is not a perfectly feature-matched one, and that must be stated before the numbers are read. The graph pipeline retained dedicated macroeconomic nodes for gold, oil, USD/INR and India VIX, whereas the reconstructed classical feature matrix did not, for the reasons given in Section 3.5. The cross-pipeline comparison in Table 5.11 should therefore be interpreted as a comparison of the two evaluated modelling pipelines as actually built, rather than as a controlled comparison of classical machine learning against graph neural networks on an identical feature set.

**Table 5.11:** Best configuration from each pipeline on comparable metrics.

| **Pipeline** | **Best configuration** | **Accuracy** | **ROC-AUC** |
|---|---|---|---|
| Classical ML | XGBoost, E2 (Technical Only) | 0.5167 | 0.5134 |
| Classical ML | Random Forest, E1 (Technical + GDELT) | 0.5077 | 0.5315 |
| Graph-based | HGT-style | 0.5168 | 0.5018 |
| Random baseline | — | $\approx$0.50 | 0.5000 |
| Constant-UP baseline (graph test partition) | — | 0.5169 | not applicable |
| Sector-wise majority-class baseline (graph test partition) | — | 0.5242 | not applicable |

Accuracy is effectively tied at 0.5167 and 0.5168, but this similarity is misleading and should not be reported without qualification: the graph figure is produced by constant-UP prediction, while the classical figure comes from a model whose probability scores provide a small amount of ranking signal above 0.50 ROC-AUC. ROC-AUC separates them, with the classical models ahead at 0.5134 and 0.5315 against 0.5018. Both baseline rows are included only for the graph partition, where the full per-sector confusion-matrix counts needed to compute them are available; the equivalent majority-class figure for the classical test partition could not be computed for the reason given in Section 5.3, so the graph accuracy of 0.5168 should not be read as sitting the same distance above a majority baseline as the classical accuracy figures do, since no classical majority baseline is available for that comparison.

The graph representation did not deliver an advantage under the conditions tested. The experiment does not establish why. The relational structure may carry little additional information beyond what the sector-technical features already encode, since a sector index and its neighbours share much of their variation and the graph may therefore add redundancy rather than signal. Alternatively, the abbreviated training may simply have prevented the models from using the structure at all. These two explanations are not distinguishable from the evidence available, and the second is the more immediately testable, which is why it heads the future scope.

**Table 5.12:** Research question outcomes.

| RQ | Question | Finding |
|---|---|---|
| RQ1 | Do GDELT-derived features improve classical prediction? | Not supported |
| RQ2 | Does any GNN beat the best classical configuration? | Not supported |
| RQ3 | Do the GNNs learn a discriminative decision boundary? | Not supported |

All three questions are answered in the negative on the evidence collected. RQ1 is negative because the GDELT features produced no reliable improvement, and the largest observed effect favoured their removal. RQ2 is negative because the best graph AUC of 0.5018 falls below both classical AUC figures. RQ3 is negative because all four graph models predicted a single class throughout the test period, as Table 5.9 documents. In each case the finding applies to the configuration evaluated, and the wording is chosen to avoid implying a more general conclusion than the evidence supports.

### *5.14 Critical Discussion*

Taken together, the observed predictive performance remained close to chance and constant-prediction reference levels across the whole study. The strongest single result across the whole study is 0.5315 AUC, roughly three points above chance, obtained by a Random Forest that was not the top-ranked model by accuracy.

This is worth stating plainly rather than presenting selectively. Reporting the highest AUC alongside the highest accuracy from a different configuration, without noting that the two orderings disagree, would give a misleading impression of consistency. Quoting the graph F1 of 0.6814 against the classical 44.23% would suggest a result that Section 5.9 shows to be an artefact of two different metric definitions. Both temptations are easy to fall into when a study is written up after the fact, and both are avoided here deliberately.

Several factors plausibly limit what was achievable in this study. Daily directional prediction has an inherently low signal-to-noise ratio, and the test period spans two years during which market conditions need not resemble the training period; a fixed model fitted on 2018 to 2022 data is being asked to generalise across a regime boundary it never observed. The graph models were trained for one to three epochs, which is a severe constraint whatever the merits of the architectures. The graph topology is fixed across the entire period, whereas the strength of a relationship such as the assumed Iran-to-Energy link plausibly varies with circumstances and may be close to irrelevant for long stretches. The classical feature set lacked dedicated macroeconomic columns, so the ablation tested GDELT on a narrower base than intended. Finally, all reported values are single-run point estimates with no variance estimate, so differences of a few tenths of a percentage point cannot be distinguished from run-to-run variation.

What the study does establish is narrower but still useful. Under a chronological protocol with training-only preprocessing, on this dataset, neither GDELT-derived features nor a domain-informed graph representation improved next-day sector direction prediction, and the graph models specifically failed to learn a discriminative boundary in the abbreviated configuration tested. The observation that a metric-definition difference can make degenerate models appear to outperform working ones is a methodological point that applies well beyond this particular experiment, and is arguably the most transferable outcome of the work.

Nothing here supports a general conclusion that graph neural networks are unsuited to financial prediction, nor that geopolitical information is irrelevant to markets. The evidence is specific to this dataset, this graph, these features, this horizon and this training configuration, and the literature reviewed in

Section 2.13 contains multiple studies reporting positive results for graph-based financial models under different designs.

# CHAPTER 6
# CONCLUSIONS AND FUTURE SCOPE

## *6.1 Major Findings*

Four findings emerge from the experiments performed.

First, the GDELT-derived geopolitical features did not improve next-day sector direction prediction. Removing them improved XGBoost on all three reported metrics and left Random Forest essentially unchanged, so the ablation provides no support for their inclusion under the tested configurations.

Second, no graph architecture outperformed the classical baselines on comparable metrics. The best graph result, 0.5018 ROC-AUC from the HGT-style model, falls below both the 0.5134 of the classical configuration ranked first by accuracy and the 0.5315 of the classical configuration ranked first by AUC.

Third, all four graph models produced constant-UP predictions on the held-out test observations, assigning UP to every test observation in every sector. Their apparent F1 advantage is an artefact of a positive-class metric definition applied to constant-UP predictions; expressed as a macro F1 derived from the reported per-sector confusion matrices, the same predictions score approximately 0.3405 against 0.4423 for the best classical configuration.

Fourth, absolute performance across the entire study remained close to chance. The highest ROC-AUC observed in any configuration was 0.5315, and the highest accuracy was 51.68%, against a constant-prediction baseline of approximately 52%.

## *6.2 Interpretation*

These results characterise a specific experimental configuration rather than the general capability of the methods involved, and three constraints in particular shape what can legitimately be concluded.

The graph models trained for one to three epochs under the abbreviated configuration, with three of the four selecting epoch 1 by validation loss. Their failure to learn a discriminative boundary is at least as consistent with insufficient training as with any property of the graph representation, and the two explanations cannot be separated from the evidence available.

The classical ablation tested GDELT-derived features against sector-technical features alone, because the reconstruction produced no dedicated macroeconomic columns. A configuration including full macroeconomic coverage might respond differently, and the ablation result should be read as applying to the feature base actually used.

All results are single-run point estimates on one chronological split without significance testing. Differences of the size reported here, a few tenths of a percentage point in several cases, cannot be distinguished from run-to-run variation on this evidence, which is why the discussion consistently avoids ranking configurations that differ only marginally.

## *6.3 Contributions*

The work contributes a domain-informed heterogeneous market graph linking sectoral, macroeconomic, country and global GDELT entities, with each relation group justified explicitly and presented as a modelling assumption; a controlled ablation that tests rather than assumes the value of geopolitical features; a four-architecture graph comparison conducted under a chronological protocol with training-only preprocessing; and a transparent account of a near-random result, including the identification of a metric-definition difference that would otherwise have inverted the apparent comparison between the two pipelines. The last of these is a negative result reported in full, which the review of related work in

Section 2.13 suggests is uncommon in this literature.

### *6.4 Limitations*

1. **GNN training and model capacity.** The GNN models were trained for a relatively short period, with a maximum of 14 epochs, and three of the four models stopped at the first epoch. The models also have a relatively large number of parameters compared with the available training data. Therefore, the reported results may not represent the full potential of the GNN architectures.

2. **Constant predictions and evaluation metrics.** All four graph models predicted UP for every test observation. This means that the models did not learn a clear distinction between UP and DOWN movements. The two pipelines also used different F1 definitions, so F1 scores were not directly comparable. Accuracy and ROC-AUC were therefore used for the main comparison between the two approaches.

3. **Data and feature limitations.** The classical pipeline did not include dedicated macroeconomic features, which means that the GDELT experiment was carried out on a more limited feature set than originally planned. Also, the exact publication times of the GDELT and macroeconomic data were not available, so their availability before the next-day prediction could not be fully verified.

4. **Evaluation limitations.** The study used one chronological train–validation–test split and each model was evaluated in a single run. As a result, the study does not provide information about how stable the results would be across different time periods or random seeds. Walk-forward validation, repeated experiments, and statistical testing could provide a more reliable assessment.

5. **Graph and feature representation.** The same graph structure was used throughout the entire study period, even though relationships between sectors and other economic variables can change over time. In addition, different node types were represented using zero-padding to create a common feature size, which may not be the most suitable representation for all node types.

6. **Explainability and practical evaluation.** The explainability analysis was based on gradient-based saliency and focused on a selected target, so it provides only a limited view of how the graph model uses its inputs. The study also did not include a trading backtest or transaction-cost analysis. Therefore, the predictive results should not be interpreted as evidence that the models could generate profitable trading strategies.

### *6.5 Conclusion*

This dissertation asked whether an explicit graph representation of Indian sectoral indices, macroeconomic variables and GDELT geopolitical entities offers an advantage over classical machine learning for next-day directional prediction, and whether geopolitical features contribute measurably. Under the evaluated dataset, feature configuration, graph topology and training protocol, the experiments did not provide evidence that the graph representation improved next-day directional prediction, nor that the evaluated GDELT features improved the classical models.

The strongest classical configuration reached 51.67% accuracy and 0.5134 ROC-AUC, while a different classical configuration reached a higher 0.5315 ROC-AUC at lower accuracy. The best graph model reached 0.5018 ROC-AUC, and all four graph models predicted a single class throughout the test period. The evaluated GDELT features produced no reliable improvement in either direction.

These outcomes are reported exactly as obtained. The graph results in particular reflect an abbreviated training configuration and should not be read as establishing the limits of these architectures; what they

establish is that the configuration tested did not work, and that the reasons are identifiable and addressable. A negative result obtained under a carefully controlled protocol is a legitimate research finding, and the methodological issue surfaced in Section 5.9, where a metric-definition difference made degenerate models appear to outperform working ones, is arguably the most transferable outcome of the study.

### *6.6 Future Scope*

The following items follow directly from the limitations listed in Section 6.4. They are proposed work and none of them has been performed in this study.

1. **Improve GNN training and model design.** Future experiments can use longer training schedules and different model settings to determine whether the poor graph results were mainly caused by insufficient training or by the graph architectures themselves.

2. **Improve the data and feature set.** The classical pipeline can be rebuilt with complete macroeconomic features, while the timing of GDELT and macroeconomic information can be checked to make sure that only information available before the prediction time is used.

3. **Use stronger evaluation methods.** Future work can use walk-forward or rolling-origin validation, repeated runs with different random seeds, and consistent evaluation metrics across both pipelines. This would give a better understanding of how stable the results are.

4. **Explore more flexible graph and temporal models.** Instead of keeping the same graph structure throughout the study, future work can investigate time-varying relationships and alternative graph structures. Temporal GNNs or recurrent models can also be explored to better capture changes over time.

5. **Improve model interpretation and practical evaluation.** Future work can extend the explainability analysis to all sectors and investigate probability calibration. If a model shows meaningful predictive performance, its usefulness can then be tested through a backtest that includes transaction costs.

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